FACTOR ANALYSIS FOR DATA ANALYSIS

A Thesis

Presented to the

Faculty of

California State Polytechnic University, Pomona

In Partial Fulfillment

Of the Requirements for the Degree

Master of Science

In

Mathematics

By

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2016

SIGNATURE PAGE

THESIS: FACTOR ANALYSIS FOR DATA ANALYSIS

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DATE SUBMITTED: Spring 2016

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ACKNOWLEDGMENTS

I would like to thank my family, who have supported me throughout my academic career. My parents have always been there to support any decision I have made and they were always willing to go the extra mile to create a comfortable and supporting place in my life. My brother has always been around for help and has granted me tenacity to push through any barriers I may run into. I would like to thank Dr. Hoon Kim for his time and allowing me the opportunity to conduct this research under his guidance. I would also like to thank Dr. Jungwon Chris Mun and Dr. Adam King for taking the time to consider my thesis. I would also like to thank Ms. Cidni Griffith and other department staff for always taking the time away from their day to help whenever I dropped by the office.

ABSTRACT

Factor analysis is a multivariate statistical tool that is used in the development, refinement and evaluation of tests and measures of data that can be used in clinical and educational contexts by an analyst. This thesis will focus on a brief understanding of the tools needed to understand and complete a factor analysis model. An example regarding consumers and their influences in buying a six-pack of beer has been analyzed and has shown that two distinct factors of either being a connoisseur of beer interested in the quality or being a cheap drunk that is searching for a way to become intoxicated.

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Chapter 1

Matrix Theory

1.1 Matrix Definitions and Examples

A *matrix* is any rectangular array of real numbers, a_{ij}. We denote an arbitrary array of n rows and p columns by

$$\mathbf{A}_{(n\times p)} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ a_{21} & a_{22} & \cdots & a_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{np} \end{bmatrix} \tag{1.1}$$

A *square matrix* is a special case of a matrix where the number of rows is equal to the number of columns, in other words, $\mathbf{A}_{(n\times p)} = \mathbf{A}_{(n\times n)}$ with $n = p$.

The *transpose operation*, $\mathbf{A}^t$, of a matrix changes the columns into rows, so that the first column of $\mathbf{A}$ becomes the first row of $\mathbf{A}^t$, the second column becomes the second row, and so forth as seen below.

$$\mathbf{A} = \begin{bmatrix} 7 & -9 & 8 \\ 9 & 5 & 6 \end{bmatrix} \qquad \mathbf{A}^t = \begin{bmatrix} 7 & 9 \\ -9 & 5 \\ 8 & 6 \end{bmatrix} \tag{1.2}$$

A square matrix is said to be *symmetric* if $\mathbf{A} = \mathbf{A}^t$ or $a_{ij} = a_{ji}$ for all i and j. The matrix

$$\begin{bmatrix} 7 & 3 \\ 3 & -8 \end{bmatrix}$$

is symmetric; the matrix

$$\begin{bmatrix} 7 & 4 \\ 6 & -8 \end{bmatrix}$$

is not symmetric.

If we let $\mathbf{I}$ denote the square matrix with ones on the diagonal and zeros elsewhere, it follows from the definition of matrix multiplication that the *(i,j)th* entry of $\mathbf{AI}$ is $a_{i1} \times 0 + \cdots + a_{i,j-1} \times 0 + a_{ij} \times 1 + a_{i,j+1} \times 0 + \cdots + a_{ik} \times 0 = a_{ij}$ so, $\mathbf{AI} = \mathbf{A}$. Similarly, $\mathbf{IA} = \mathbf{A}$, so $\mathbf{I}_{k\times k}\mathbf{A}_{k\times k} = \mathbf{A}_{k\times k}\mathbf{I}_{k\times k} = \mathbf{A}_{k\times k}$.

This works for any square matrix $\mathbf{A}_{k\times k}$, so the matrix $\mathbf{I}$ acts like 1 in ordinary multiplication ($1 \cdot a = a \cdot 1 = a$), so it is called the *identity matrix*.

The fundamental scalar relation about the existence of an inverse number, a^{-1}, such that $a^{-1}a = aa^{-1} = 1$ if $a \neq 0$ has the following matrix algebra extension: If there exists a matrix $\mathbf{B}$ such that $\mathbf{B}_{k\times k}\mathbf{A}_{k\times k} = \mathbf{A}_{k\times k}\mathbf{B}_{k\times k} = \mathbf{I}_{k\times k}$ then $\mathbf{B}$ is called the *inverse of* $\mathbf{A}$ and is denoted by $\mathbf{A}^{-1}$.

An array $\mathbf{x}$ of n real numbers $x_1, x_2, \ldots, x_n$ is called a vector, and it is written as

$$\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \qquad \mathbf{x}^t = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \tag{1.3}$$

where the t denotes the operation of transposing a column to a row.

A matrix may also be multiplied by a constant c. The product $c\mathbf{A}$ is the matrix that results from multiplying each element of $\mathbf{A}$ by c. Thus

$$c\mathbf{A}_{(n\times p)} = \begin{bmatrix} ca_{11} & ca_{12} & \cdots & ca_{1p} \\ ca_{21} & ca_{22} & \cdots & ca_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{n1} & ca_{n2} & \cdots & ca_{np} \end{bmatrix} \tag{1.4}$$

Two matrices of the same dimensions can be added. If $\mathbf{A}_{(n\times p)} + \mathbf{B}_{(n\times p)} = \mathbf{C}_{(n\times p)}$, then $c_{ij} = a_{ij} + b_{ij}$ for any (i, j). Let

$$\mathbf{A} = \begin{bmatrix} 10 & 7 & 9 \\ 9 & -9 & 9 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 9 & -8 & -7 \\ 8 & 5 & 9 \end{bmatrix}$$

Then

$$\mathbf{C} = \begin{bmatrix} 10+9 & 7-8 & 9-7 \\ 9+8 & -9+5 & 9+9 \end{bmatrix} = \begin{bmatrix} 19 & -1 & 2 \\ 17 & -4 & 18 \end{bmatrix} \qquad 4\mathbf{A} = \begin{bmatrix} 40 & 28 & 36 \\ 36 & -36 & 36 \end{bmatrix}$$

It is also possible to define the multiplication of two matrices if the dimensions of the matrices conform in the following manner: When $\mathbf{A}$ is $(n\times k)$ and $\mathbf{B}$ is $(k\times p)$, so that the number elements in a row of $\mathbf{A}$ is the same as the number elements in a column of $\mathbf{B}$, we can form the matrix product $\mathbf{AB}$. An element of the new matrix $\mathbf{AB}$ is formed by taking the inner product of each row of $\mathbf{A}$ with each column of $\mathbf{B}$. Providing the matrix product $\mathbf{A}_{(n\times k)}\mathbf{B}_{(k\times p)} = \mathbf{AB}_{(n\times p)} = \mathbf{D}_{(n\times p)}$, where $d_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots + a_{ik}b_{kj} = \sum_{l=1}^{k} a_{il}b_{lj}$.

Let

$$\mathbf{A} = \begin{bmatrix} 2 & -3 & 0 \\ -1 & 3 & 2 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} -4 \\ 5 \\ 7 \end{bmatrix} \qquad \mathbf{C} = \begin{bmatrix} 0 & -2 \\ -1 & -3 \end{bmatrix}$$

Then

$$\mathbf{A}_{(2\times3)}\mathbf{B}_{(3\times1)} = \begin{bmatrix} 2(-4)+-3(5)+0(7) \\ -1(-4)+3(5)+2(7) \end{bmatrix} = \begin{bmatrix} -23 \\ 33 \end{bmatrix}_{(2\times1)}$$

$$\mathbf{C}_{(2\times2)}\mathbf{A}_{(2\times3)} = \begin{bmatrix} 0(2)+-2(-1) & 0(-3)+-2(3) & 0(0)+-2(2) \\ -1(2)+-3(-1) & -1(-3)+-3(3) & -1(0)+-3(2) \end{bmatrix} = \begin{bmatrix} 2 & -6 & -4 \\ 1 & -6 & -6 \end{bmatrix}_{(2\times3)}$$

A square matrix $\mathbf{A}_{k\times k}$ is *nonsingular* if $\mathbf{A}_{k\times k}\mathbf{x}_{k\times1} = \mathbf{0}_{k\times1}$ implies that $\mathbf{x}_{k\times1} = \mathbf{0}_{k\times1}$. If a matrix fails to be nonsingular, it is called *singular*. Equivalently, a square matrix is nonsingular if its rank, the number of linearly independent rows (or columns), is equal to the number of rows (or columns) it has.

Given a vector and a matrix, the *quadratic form* is given as $\mathbf{y}^t\mathbf{A}\mathbf{y} = \sum_{i=1}^{k} y_i\mathbf{A}y_i = \sum_{i=1}^{k}\sum_{j=1}^{k} a_{ij}y_iy_j$.

$\mathbf{A}$ is said to be *positive definite* if $\mathbf{A} = \mathbf{A}^t$ and $\mathbf{y}^t\mathbf{A}\mathbf{y} > 0$, $\forall\ \mathbf{y} \in \Re^k$ where $\mathbf{y} = \mathbf{0}$. $\mathbf{A}$ is said to be *positive semi definite* if $\mathbf{A} = \mathbf{A}^t$ and $\mathbf{y}^t\mathbf{A}\mathbf{y} \geq 0$, $\forall\ \mathbf{y} \in \Re^k$.

Eigenvalues are a special set of scalars associated with a matrix equation. For a matrix $\mathbf{A}_{k\times k}$, if there is a vector $\mathbf{x}$ and a scalar λ such that $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$, where $\mathbf{x} = \mathbf{0}$, then λ is called the eigenvalue of $\mathbf{A}$ with corresponding eigenvector $\mathbf{x}$.

The *spectral decomposition* can be seen when a symmetric matrix $\mathbf{A}_{k\times k}$ can be fac-

tored in terms of its eigenvalues and eigenvector pairs as

$$\begin{aligned}\mathbf{A} &= \lambda_1 \mathbf{e}_1 {\mathbf{e}_1}^t + \lambda_2 \mathbf{e}_2 {\mathbf{e}_2}^t + \cdots + \lambda_p \mathbf{e}_p {\mathbf{e}_p}^t \\ &= \sum_{i=1}^{k} \lambda_i \mathbf{e}_i {\mathbf{e}_i}^t\end{aligned}$$

The vector $\mathbf{y} = a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \cdots + a_k\mathbf{x}_k$ is a *linear combination* of the vectors $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_k$. A set of vectors $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_k$ is said to be *linearly dependent* if there exist k numbers $(a_1, a_2, \cdots, a_k)$, not all zero, such that

$$a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \cdots + a_k\mathbf{x}_k = \mathbf{0}$$

otherwise the set of vectors is said to be *linearly independent*.

1.2 Matrix Representation of a Data Set

Descriptive statistics computed from n measurements on p variables can also be organized into data arrays. Let $x_{11}, x_{21}, \cdots, x_{n1}$ be n measurements on the first variable, then the average of these measurements is

$$\bar{x}_1 = \frac{1}{n}\sum_{j=1}^{n} x_{j1}$$

This is the *sample mean* for the first variable, $\bar{x}_1$. Given p variables the sample means can be arranged into a data array such as

$$\bar{x} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_p \end{bmatrix}$$

A measure of spread is provided by the *sample variance* defined for n measurements on the first variable as

$$s_1^2 = s_{11} = \frac{1}{n}\sum_{j=1}^{n}(x_{j1} - \bar{x}_1)^2$$

in general the sample variances for p variables can be represented as the following.

$$s_k^2 = s_{kk} = \frac{1}{n}\sum_{j=1}^{n}(x_{jk} - \bar{x}_k)^2, \qquad k = 1, 2, \cdots, p$$

Consider n pairs of measurements on each of variables 1 and 2,

$$\begin{bmatrix} x_{11} \\ x_{12} \end{bmatrix}, \begin{bmatrix} x_{21} \\ x_{22} \end{bmatrix}, \cdots, \begin{bmatrix} x_{n1} \\ x_{n2} \end{bmatrix}$$

that is x_{j1} and x_{j2} are observed on the jth experimental item, where $j = 1, 2, \cdots, n$. A measure of linear association between measurements of variables 1 and 2 is provided by the sample covariance

$$s_{12} = \frac{1}{n}\sum_{j=1}^{n}(x_{j1} - \bar{x}_1)(x_{j2} - \bar{x}_2)$$

or the average product of the deviations from their respective means. In general the *sample covariance* between any two variables, i and k, is as follows.

$$s_{ik} = \frac{1}{n}\sum_{j=1}^{n}(x_{ji}-\bar{x}_i)(x_{jk}-\bar{x}_k)$$

It should be noted that when $i = k$ the sample covariance reduces to the sample variance. Moreover, $s_{ik} = s_{ki}$ for all i and k. These sample variances and covariances can be arranged into the following data array.

$$\mathbf{S}_n = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1p} \\ s_{21} & s_{22} & \cdots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{p1} & s_{p2} & \cdots & s_{pp} \end{bmatrix}$$

The *sample correlation coefficient* is the measure of the linear association between two variables regardless of unit of measurement. This sample correlation coefficient for the ith and k^{th} variables is defined as

$$\begin{aligned} r_{ik} &= \frac{s_{ik}}{\sqrt{s_{ii}}\sqrt{s_{kk}}} \\ &= \frac{\sum_{j=1}^{n}(x_{ji}-\bar{x}_i)(x_{jk}-\bar{x}_k)}{\sqrt{\sum_{j=1}^{n}(x_{ji}-\bar{x}_i)^2}\sqrt{\sum_{j=1}^{n}(x_{jk}-\bar{x}_k)^2}} \end{aligned}$$

for $i = 1, 2, \cdots, p$ and $k = 1, 2, \cdots, p$. As with the sample covariances the sample correlations share the property that $r_{ik} = r_{ki}$ for all i and k. This sample correlation coefficient becomes a standardized version of the sample covariance. As with the sample means and sample covariances, the sample correlations for p variables can be set up in the following

data array.

$$\mathbf{R} = \begin{bmatrix} 1 & r_{12} & \cdots & r_{1p} \\ r_{21} & 1 & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p1} & r_{p2} & \cdots & 1 \end{bmatrix}$$

Suppose the original values x_{ij} and x_{jk} are replaced by *standardized* values $(x_{ji} - \bar{x}_i)/\sqrt{s_{ii}}$ and $(x_{jk} - \bar{x}_k)/\sqrt{s_{kk}}$. These new standardized values are commensurable as both values have been centered about zero and expressed through standard deviation units. The sample correlation coefficients simply are the sample covariance using standardized observations in place of the original values.

Chapter 2

Regression Models

2.1 Linear Regression Model

Regression analysis is a **statistical methodology** for predicting values of one or more *response* (dependent) variables from a collection of *predictor* (independent) variables. It can also be used for assessing the effects of the predictor variables on the responses. The name *regression* was developed by F. Galton (1822-1911) during his first paper "Kinship and Correlation" on the subject reflecting the data for which he was working.

Three major principles for regression analysis are as follows: *Data Description, Control*, and *Prediction*.

- *Data Description* : The regression model is a tool to provide a summary and explanation of a data set.
- *Control* : The regression model can minimize variable collection.
- *Prediction* : Response variables can be predicted based on regression models.

2.1.1 Simple Linear Regression

A *simple linear regression model* contains a response **Y** that can be explained by a linear function of the regressor **X**. The simple linear regression model is as follows.

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon$$

with

$$\mathbf{Y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

where

- **Y** is a vector of response variable.
- **X** is a matrix of regressor variable.
- ε is a vector of random error components. These error terms are assumed to be independent and normally distributed with mean zero and variance $\sigma^2\mathbf{I}$.
- β_0 and β_1 are known as the *regression coefficients*. These are the unknown parameters to be estimated in order to predict the response **Y** for given values of the regressor **X**.

The regressor **X** is treated as an independent variable, due to being recorded from an experiment or set by an investigator, and the response **Y** as the dependent variable, due to the presence of random error components. The mean and variance of **Y** can be calculated as follows.

$$
\begin{aligned}
E(\mathbf{Y}|\mathbf{X}) &= E(\mathbf{X}\beta + \varepsilon|\mathbf{X}) \\
&= E(\mathbf{X}\beta) + E(\varepsilon) \\
&= \mathbf{X}\beta + \mathbf{0} \\
&= \mathbf{X}\beta
\end{aligned}
$$

with

$$
\begin{aligned}
Var(\mathbf{Y}|\mathbf{X}) &= Var(\mathbf{X}\beta + \varepsilon) \\
&= Var(\mathbf{X}\beta) + Var(\varepsilon) \\
&= 0 + Var(\varepsilon) \\
&= Var(\varepsilon) \\
&= \sigma^2\mathbf{I}
\end{aligned}
$$

Estimation of the parameters

Given n pairs of data, $(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)$, the simple regression model would change as follows.

$$
y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad\qquad i = 1, 2, \cdots, n
$$

The least-squares criterion selects β_0 and β_1 so as to minimize the sum of the squares of the error, $\varepsilon_i = y_i - \beta_0 - \beta_1 x_i$, with $i = 1, 2, \cdots, n$.

$$
S(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2
$$

The least-squares estimators of β_0 and β_1, denoted as $\hat{\beta}_0$ and $\hat{\beta}_1$, can be obtained by minimizing the least-squares criterion as follows.

$$
\frac{\partial S(\beta_0, \beta_1)}{\partial \beta_0} = -2 \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0
$$

and

$$\frac{\partial S(\beta_0, \beta_1)}{\partial \beta_1} = -2\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)x_i = 0$$

Simplifying these two partial derivative equations.

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i$$

$$\hat{\beta}_0 \sum_{i=1}^{n} x_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} y_i x_i$$

These simplified equations are called the *least-squares normal equations*. Solving this pair of equations provides the following solutions.

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

and

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i - \frac{1}{n}\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{\sum_{i=1}^{n} x_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n} x_i\right)^2} \tag{2.1}$$

where

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \quad \text{and} \quad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

are the means of y_i and x_i, respectively. The coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$ are the least-squares estimators of β_0 and β_1. Therefore, the simple linear regression model can now be written as

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Simplifying Equation 2.1 to

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

where

$$S_{xy} = \sum_{i=1}^{n}(y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^{n} y_i x_i - \frac{1}{n} \quad \sum_{i=1}^{n} y_i \qquad \sum_{i=1}^{n} x_i \quad = \sum_{i=1}^{n} y_i x_i - n\bar{y}\bar{x}$$

and

$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{1}{n} \quad \sum_{i=1}^{n} x_i \quad {}^{2} = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

In order to test the adequacy of this model a residual analysis needs to be performed. The residuals can be found using the difference between the observed response y_i and the corresponding fitted value $\hat{y}_i$. The formula for the residual e_i is as follows.

$$e_i = y_i - \hat{y}_i, \qquad i = 1, 2, \cdots, n.$$

The Analysis of Variance

Analysis of Variance (ANOVA) is a common tool to analyze the significance of these regression values, to test variabilities within the regression model, and to compare the differences between group means. The total variability of the observation is the *total*

sum of squares, *SST*, using a partitioning as follows.

$$
\begin{aligned}
\sum_{i=1}^{n}(y_i-\bar{y})^2 &= \sum_{i=1}^{n}[(\hat{y}_i-\bar{y})+(y_i-\hat{y}_i)]^2 \\
&= \sum_{i=1}^{n}[(\hat{y}_i-\bar{y})^2+(y_i-\hat{y}_i)^2+2(\hat{y}_i-\bar{y})(y_i-\hat{y}_i)] \\
&= \sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2+\sum_{i=1}^{n}(y_i-\hat{y}_i)^2+2\sum_{i=1}^{n}(\hat{y}_i-\bar{y})(y_i-\hat{y}_i) \\
&= \sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2+\sum_{i=1}^{n}(y_i-\hat{y}_i)^2+2\sum_{i=1}^{n}[\hat{y}_i(y_i-\hat{y}_i)-\bar{y}(y_i-\hat{y}_i)] \\
&= \sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2+\sum_{i=1}^{n}(y_i-\hat{y}_i)^2+2\sum_{i=1}^{n}\hat{y}_i(y_i-\hat{y}_i)-2\sum_{i=1}^{n}\bar{y}(y_i-\hat{y}_i) \\
&= \sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2+\sum_{i=1}^{n}(y_i-\hat{y}_i)^2+2\sum_{i=1}^{n}\hat{y}_i e_i-2\bar{y}\sum_{i=1}^{n}e_i \\
&= \sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2+\sum_{i=1}^{n}(y_i-\hat{y}_i)^2+0+0 \qquad \text{(by properties of least-squares estimators)} \\
&= \sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2+\sum_{i=1}^{n}(y_i-\hat{y}_i)^2
\end{aligned}
$$

The SST has $df_T = n-1$ total degrees of freedom. The first term on the right hand side is the *regression sum of squares*, *SSR*, which represents the variability in the observation y_i and has $df_R = 1$ model degree of freedom. The second term on the right hand side is the *residual sum of squares*, *SSE*, which represents the error variations provided by the model and has $df_E = n-2$ error degrees of freedom. This partitioning can be represented as follows.

$$SST = SSR + SSE$$

Furthermore, with $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

$$\begin{aligned} SSR &= \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 \\ &= \sum_{i=1}^{n} (\hat{\beta}_0 + \hat{\beta}_1 x_i - \hat{\beta}_0 - \hat{\beta}_1 \bar{x})^2 \\ &= (\hat{\beta}_1)^2 \sum_{i=1}^{n} (x_i - \bar{x})^2 \\ &= \hat{\beta}_1{}^2 S_{xx} \\ &= \hat{\beta}_1 S_{xy} \quad \text{with} \quad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \end{aligned}$$

While using ANOVA to analyze the significance of the linear regression model, the F-distribution is used because SSR and SSE are independent and each follows χ^2 distribution with $df_R = 1$ and $df_E = n-2$. This creates an F test under null hypothesis of the slope parameter is zero, $H_0 : \beta_1 = 0$.

Using the F test, the following test statistic is provided

$$F = \frac{SSR/df_R}{SSE/df_E} = \frac{SSR/1}{SSE/(n-2)} = \frac{MSR}{MSE}$$

where MSR is the regression mean-squared value and MSE is the residual mean-squared value.

This provides the following ANOVA table as in Table 2.1.

Table 2.1: Analysis of Variance for Testing Significance of Regression

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares	F
Regression	$SSR = \hat{\beta}_1 S_{xy}$	1	MSR	MSR/MSE
Error	$SSE = SST - \hat{\beta}_1 S_{xy}$	$n-2$	MSE	
Total	SST	$n-1$		

Coefficient of Determination

The *coefficient of determination,* R^2*,* is a way to assess the overall accuracy of the linear regression model that represents the percentage of variance in the response that can be explained by the regressor. R^2 can be defined as follows.

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

As the value of R^2 approaches 1, most of the variability in the response variable y_i is being explained by the regressor x_i for a regression model $\hat{y}_i$.

While the value of R^2 helps determine the accuracy of the linear regression model, the value of R^2 can be increased by adding more predictive variables, irrespective of how useful they are to the response variable, into the model. Thus, using an *adjusted* R^2 is more desirable as it will only increase on adding a variable to the model if the addition of the variable reduces the MSE. The adjusted R^2 can be defined as follows.

$$R^2_{adj} = 1 - \frac{SSE/df_E}{SST/df_T} = 1 - \frac{MSE}{SST/df_T}$$

2.1.2 Multiple Linear Regression

A *multiple regression model* is a linear regression model involving more than one predictor variable. In general, the response y_i may involve up to p predictor variables and this multiple regression model can be expressed as follows.

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \varepsilon_i \tag{2.2}$$

where

- ε_i are random error terms that are assumed to be independent and normally distributed with mean zero and some variance σ^2.
- β_i, for $i = 0, 1, \cdots, p$, are known as the regression coefficients. These fixed unknown coefficients are to be estimated.

While the regression coefficients must follow a linear trend, the predictor variables do not have any restrictions. Some models that may be more complicated than Equation 2.2 can also considered as a multiple linear regression model because the restriction is limited to the linearity in regression coefficients. Some examples of more complicated multiple linear regression models are as follows:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2^2 + \varepsilon$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \varepsilon$$

$$log(y) = \beta_0 + \beta_1 \sin(x_1) + \beta_2 \ln(x_2^2) + \varepsilon$$

In matrix notation, with n observations and $k(= p+1)$ parameters, a multiple linear regression model can be represented as

$$\mathbf{Y}_{(n\times 1)} = \mathbf{X}_{(n\times k)} \beta_{(k\times 1)} + \varepsilon_{(n\times 1)} \tag{2.3}$$

where

- $\mathbf{Y}$ is a vector of responses, with mean $E(\mathbf{Y}) = \mathbf{X}\beta$.
- $\mathbf{X}$ is a model design matrix for the fixed effects of predictor variables in group i.
- β is a vector of regression coefficients.
- ε is the *residual* vector, a vector of error terms.

The least-square estimators, $\hat{\beta}$, for the vector β can be obtained by minimizing

$$S(\beta) = \sum_{i=1}^{n} \varepsilon_i^2 = \varepsilon^t \varepsilon = (\mathbf{Y} - \mathbf{X}\beta)^t (\mathbf{Y} - \mathbf{X}\beta)$$

After simplification the vector form of fitted multiple linear regression model using the *hat matrix* $\mathbf{H}_{(n\times n)} = \mathbf{X}(\mathbf{X}^t\mathbf{X})^{-1}\mathbf{X}^t$:

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}^t\mathbf{X})^{-1}\mathbf{X}^t\mathbf{Y} = \mathbf{HY} \quad \text{where} \quad \hat{\beta} = (\mathbf{X}^t\mathbf{X})^{-1}\mathbf{X}^t\mathbf{Y}$$

This provides the residual vector $\mathbf{e}$ and the adequacy of the model can be tested through a residual analysis similar to the simple linear regression. The residual vector is as follows:

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X}\hat{\beta} = \mathbf{Y} - \mathbf{HY} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$$

After the initial set up of all the vectors and matrices the ANOVA process can be used to analyze the regression values produced. The process works similar to the simple linear regression as in Table 2.1, but the degrees of freedom change to compensate for the added variables $df_R = k - 1$, $df_E = n - k$, and $df_T = n - 1$. Then the coefficient of determination can be used once again to see if the regression model adequately explains the variability of the response variable, in this case the adjusted R^2 is more desirable.

Chapter 3

Principal Component Analysis

3.1 Principal Component Analysis

Principal component analysis, **PCA**, is a statistical tool used in exploratory data analysis for data interpretation as well as dimension reduction in order to explain the variance-covariance structure of sets of variables. In other words, principal component analysis attempts to convert a set of p possibly correlated variables into a smaller set of k linearly uncorrelated combinations of variables known as *principal components* in order to reduce the dimensionality of multivariate data while preserving as much relevant information as possible. The number of principal components, k, is less than or equal to the number of original variables, p. Principal components should be used more as a tool to understand and compact data as the components found could be used in multiple regression, cluster analysis, or even act as an intermediate step in factor analysis.

Let the random vector $\mathbf{X}^t = [x_1, x_2, \cdots, x_p]$ have the covariance matrix Σ. Consider the principal components $\mathbf{Y} = [y_1, y_2, \cdots, y_p]$, with each as a linear combination of the p

random variables $x_1, x_2, \cdots, x_p$ as follows.

$$
\begin{aligned}
y_1 &= \mathbf{a}_1^t\mathbf{x} = a_{11}x_1 + a_{12}x_2 + \cdots + a_{1p}x_p \\
y_2 &= \mathbf{a}_2^t\mathbf{x} = a_{21}x_1 + a_{22}x_2 + \cdots + a_{2p}x_p \\
\vdots &= \vdots = \qquad\qquad\qquad \vdots \\
y_p &= \mathbf{a}_p^t\mathbf{x} = a_{p1}x_1 + a_{p2}x_2 + \cdots + a_{pp}x_p
\end{aligned}
$$

Furthermore,

$$
\begin{aligned}
Var(y_i) &= \mathbf{a}_i^t\Sigma\mathbf{a}_i, \qquad i = 1, 2, \cdots, p \\
Cov(y_i, y_j) &= \mathbf{a}_i^t\Sigma\mathbf{a}_j, \qquad i, j = 1, 2, \cdots, p
\end{aligned}
$$

Taking a closer look at the first principal component it would like to maximize the variance, $Var(y_1) = \mathbf{a}_1^t\Sigma\mathbf{a}_1$ in such a way that $\mathbf{a}_1$ can increase by being multiplied by some constant, but to rule out indeterminacy it is commonplace to restrict the coefficient vectors to unit length. Then defining the first principal component with the idea that it is a linear combination $\mathbf{a}_1^t\mathbf{x}$ that maximizes $Var(\mathbf{a}_1^t\mathbf{x})$ subject to $\mathbf{a}_1^t\mathbf{a}_1 = 1$. Followed by the second principal component with the linear combination $\mathbf{a}_2^t\mathbf{x}$ that maximizes $Var(\mathbf{a}_2^t\mathbf{x})$ subject to $\mathbf{a}_2^t\mathbf{a}_2 = 1$ and $Cov(\mathbf{a}_1^t\mathbf{x}, \mathbf{a}_2^t\mathbf{x}) = 0$. This is continued until the ith step we have a linear combination $\mathbf{a}_i^t\mathbf{x}$ that maximizes $Var(\mathbf{a}_i^t\mathbf{x})$ subject to $\mathbf{a}_i^t\mathbf{a}_i = 1$ and $Cov(\mathbf{a}_i^t\mathbf{x}, \mathbf{a}_j^t\mathbf{x}) = 0$ for $j < i$.

Let Σ be the covariance matrix associated with the random vector $\mathbf{x}^t = [x_1, x_2, \cdots, x_p]$ with Σ containing the eigenvalue eigenvector pairs $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \cdots, (\lambda_p, \mathbf{e}_p)$ such that $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$. Then the ith principal component becomes

$$
y_i = \mathbf{e}_i^t\mathbf{x} = e_{i1}x_1 + e_{i2}x_2 + \cdots + e_{ip}x_p, \qquad i = 1, 2, \cdots, p
$$

and with these choices the variance and covariance of these components become

$$Var(y_i) = \mathbf{e}_i^t \Sigma \mathbf{e}_i = \lambda_i, \qquad i = 1, 2, \cdots, p$$

$$Cov(y_i, y_j) = \mathbf{e}_i^t \Sigma \mathbf{e}_j = 0, \qquad i = j$$

Therefore it results that the principal components are uncorrelated and have variances equal to the eigenvalues of Σ. Furthermore the total population variance, $\sigma_{11} + \sigma_{22} + \cdots + \sigma_{pp}$ is equal to the sum $\lambda_1 + \lambda_2 + \cdots + \lambda_p$ of all the eigenvalues of Σ. A consequence of this is the proportion of total population variance due to the kth principal component

$$\frac{\lambda_k}{\lambda_1 + \lambda_2 + \cdots + \lambda_p}, \qquad k = 1, 2, \cdots, p$$

If *most* of the total population variance, for large p, can be attributed to the first one, two, or three components, then these components can be used in place of the original p variables without much loss of information. Additionally each component of the coefficient vector $\mathbf{e}_i^t = [e_{i1}, \cdots, e_{ik}, \cdots, e_{ip}]$ is important to inspect as the magnitude of e_{ik} measures the importance of the kth variable to the ith principal component regardless of the other variables.

Although principal components are not invariant therefore it is a possibility to use standardized variables by centering $\mathbf{x}$, in which case

$$z_i = \frac{x_i - \mu_i}{\sqrt{\sigma_{ii}}}, \qquad \text{for} \quad i = 1, \cdots, p$$

and the covariance matrix, Σ, that is used with the random vector $\mathbf{x}$ to find the principal components through eigenvalues and eigenvectors becomes the eigenvalues and eigenvectors of the correlation matrix, ρ, of the random vector $\mathbf{x}$. With these new standardized variables the proportion of total variance explained due to the kth principal component of $\mathbf{z}$ is

$$\frac{\lambda_k}{p}, \qquad k = 1, 2, \cdots, p$$

where λ_k's are the eigenvalues of the correlation matrix ρ.

Chapter 4

Factor Analysis

4.1 Objectives of Factor Analysis

The objective of *factor analysis*, **FA**, is to describe the covariance relationships among several variables in terms of a few underlying, unobservable, variables called *factors*. These factors will then be used to see if the data is consistent with a predetermined structure. While factor analysis is often considered an extension of principal component analysis because both attempt to approximate the covariance matrix, the approximation found through factor analysis is more elaborate. The factor model is motivated when assessing variables by their correlation and when all variables within a grouping have a high correlation among themselves, but relatively small correlations with variables in other groupings then there is the idea that an underlying factor is responsible for these observed correlations. The factor analyst would interpret each factor according to the meaning of the variables within the grouping.

4.2 The Orthogonal Factor Model

The random vector $\mathbf{x}$, with p components, mean μ, and covariance matrix Σ put through the factor model states that $\mathbf{x}$ is linearly dependent upon a few unobservable random variables $F_1, F_2, \cdots, F_m$ called common factors, with p additional sources of variation, errors, $\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_p$. The model is as follows.

$$
\begin{aligned}
x_1 - \mu_1 &= \ell_{11}F_1 + \ell_{12}F_2 + \cdots + \ell_{1m}F_m + \varepsilon_1 \\
x_2 - \mu_2 &= \ell_{21}F_1 + \ell_{22}F_2 + \cdots + \ell_{2m}F_m + \varepsilon_2 \\
\vdots \qquad & \qquad\qquad \vdots \qquad\qquad\qquad \vdots \\
x_p - \mu_p &= \ell_{p1}F_1 + \ell_{p2}F_2 + \cdots + \ell_{pm}F_m + \varepsilon_p
\end{aligned}
$$

which can be rewritten in matrix notation as

$$
\begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \\ \vdots \\ x_p - \mu_p \end{bmatrix} = \begin{bmatrix} \ell_{11} & \ell_{12} & \cdots & \ell_{1m} \\ \ell_{21} & \ell_{22} & \cdots & \ell_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{p1} & \ell_{p2} & \cdots & \ell_{pm} \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ \vdots \\ F_m \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_p \end{bmatrix}
$$

or equivalently $(\mathbf{x} - \mu)_{p\times 1} = \mathbf{L}_{p\times m}\mathbf{F}_{m\times 1} + \varepsilon_{p\times 1}$ where

- μ_i is the mean of variable $\mathbf{x}_i$.
- ℓ_{ij} is called the loading of the ith variable on the jth factor with $\mathbf{L}$ called the matrix of factor loading.
- F_j is the jth common factor.
- ε_i are the *errors* or *specific factors* associated with the ith response x_i.

In this model the p deviations $x_1-\mu_1, x_2-\mu_2, \cdots, x_p-\mu_p$ are expressed in terms of $p+m$ random variables $F_1, F_2, \cdots, F_m, \varepsilon_1, \varepsilon_2, \cdots, \varepsilon_p$. These random variables are unobservable and this is one of the key differences between the factor model and the multiple linear regression model as the independent variable, in this case $\mathbf{F}$, is observable in the multiple linear regression model.

The Orthogonal Factor Model Assumptions

By construction there is an assumption of linearity in $\mathbf{F}$ in these factor analysis models. Also we assume that

$$E(\mathbf{F}) = \mathbf{0}_{(m\times 1)}, \qquad Cov(\mathbf{F}) = E[\mathbf{F}\mathbf{F}^t] = \mathbf{I}_{(m\times m)}$$

$$E(\varepsilon) = \mathbf{0}_{(p\times 1)}, \qquad Cov(\varepsilon) = E[\varepsilon\varepsilon^t] = \Psi_{(p\times p)} = \begin{bmatrix} \psi_1 & 0 & \cdots & 0 \\ 0 & \psi_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \psi_p \end{bmatrix}$$

and that $\mathbf{F}$ and ε are independent, so

$$Cov(\varepsilon,\mathbf{F}) = E[\varepsilon\mathbf{F}^t] = \mathbf{0}_{(p\times m)}$$

This provides the following.

$$\begin{aligned}
\Sigma = Cov(\mathbf{x}) &= E[(\mathbf{x}-\mu)(\mathbf{x}-\mu)^t] \\
&= E[(\mathbf{LF}+\varepsilon)(\mathbf{LF}+\varepsilon)^t] \\
&= E[\mathbf{LF}(\mathbf{LF})^t + \varepsilon(\mathbf{LF})^t + \mathbf{LF}(\varepsilon)^t + \varepsilon(\varepsilon)^t] \\
&= \mathbf{L}E(\mathbf{F}\mathbf{F}^t)\mathbf{L}^t + E(\varepsilon\mathbf{F}^t)\mathbf{L}^t + \mathbf{L}E(\mathbf{F}\varepsilon^t) + E(\varepsilon\varepsilon^t) \\
&= \mathbf{LIL}^t + \mathbf{0L}^t + \mathbf{L0}^t + \Psi \\
&= \mathbf{LL}^t + \Psi
\end{aligned}$$

Furthermore,

$$\begin{aligned}
Cov(\mathbf{x},\mathbf{F}) &= E\left[(\mathbf{x}-\mu)\mathbf{F}^t\right] \\
&= E\left[(\mathbf{LF}+\varepsilon)\mathbf{F}^t\right] \\
&= E(\mathbf{LFF}^t)+E(\varepsilon\mathbf{F}^t) \\
&= \mathbf{L}E(\mathbf{FF}^t)+E(\varepsilon\mathbf{F}^t) \\
&= \mathbf{LI}+\mathbf{0} \\
&= \mathbf{L}
\end{aligned}$$

and we have

$$Var(x_i) = \sigma_{ii} = {\ell_{i1}}^2 + \cdots + {\ell_{im}}^2 + \psi_i$$

The ith *communality* is the portion of the variance, $Var(x_i) = \sigma_{ii}$, of the ith variable contributed by the m common factors. The remaining portion of the variance, ψ_i, is due to the specific factor and is called the *uniqueness*, or *specific variance*. Denoting the ith communality by ${h_i}^2$ the variance can be boiled down to

$$\begin{aligned}
Var(x_i) = \sigma_{ii} &= {\ell_{i1}}^2 + \cdots + {\ell_{im}}^2 + \psi_i \\
&= {h_i}^2 + \psi_i, \qquad \text{with} \qquad i = 1,2,\cdots,p
\end{aligned}$$

which provides that the ith communality is the sum of squares of the loadings of the ith variable on the m common factors. When m is relatively small when compared to p factor analysis is most useful, and in this case the factor model is able to provide a simple explanation of the covariation in $\mathbf{X}$ with fewer parameters than the original $p(p+1)/2$ parameters in Σ. For instance, in a case where $\mathbf{X}$ contains $p = 15$ variables the orthogonal factor model, with $m = 2$, provides a Σ containing $p(p+1)/2 = 120$ parameters whereas this could also be described in terms of parameters of mp factor loadings, ℓ_{ij}, and p specific variances, ψ_i, in this case $mp + p = 45$ which is considerably fewer parameters.

While the case where the number of factors m is considerably less than the number of variables p provides a simple explanation of the covariation in $\mathbf{X}$ with fewer parameters than the parameters in Σ, most covariance matrices cannot be factored as $\mathbf{L}\mathbf{L}^t + \Psi$. Although whenever $m > 1$ there is some inherent ambiguity in the factor model. This can be seen when letting $\mathbf{T}$ be any $m \times m$ orthogonal matrix such that $\mathbf{T}\mathbf{T}^t = \mathbf{T}^t\mathbf{T} = \mathbf{I}$ then the orthogonal factor model can be rewritten as the following.

$$\begin{aligned}(\mathbf{x} - \mu) &= \mathbf{L}\mathbf{F} + \varepsilon \\ &= \mathbf{L}\mathbf{T}\mathbf{T}^t\mathbf{F} + \varepsilon \\ &= \mathbf{L}^*\mathbf{F}^* + \varepsilon, \qquad \text{where} \qquad \mathbf{L}^* = \mathbf{L}\mathbf{T} \quad \text{and} \quad \mathbf{F}^* = \mathbf{T}^t\mathbf{F}\end{aligned}$$

Since

$$E(\mathbf{F}^*) = \mathbf{T}^t E(\mathbf{F}) = \mathbf{0}$$

and

$$Cov(\mathbf{F}^*) = \mathbf{T}^t Cov(\mathbf{F})\mathbf{T} = \mathbf{T}^t\mathbf{I}\mathbf{T} = \mathbf{T}^t\mathbf{T} = \mathbf{I}$$

satisfy all the previous orthogonal factor model assumptions it is not possible, based on the observations on $\mathbf{x}$, to distinguish the loadings $\mathbf{L}$ from the loadings $\mathbf{L}^*$. Furthermore the communalities are also unaffected by the choice of $\mathbf{T}$, which provides an ambiguity that allows for the factor analyst to apply a "factor rotation" as the orthogonal matrices corresponding to rotations of the coordinate system for $\mathbf{x}$.

4.3 Methods of Estimation

The sample covariance matrix $\mathbf{S}$ is an estimator of the unknown population covariance matrix Σ. If the off-diagonal elements of $\mathbf{S}$ are small or those of the sample correlation matrix $\mathbf{R}$ are essentially zero then a factor analysis will not be useful as the specific factors play the dominant role in these situations and the factor analyst's goal is to determine a few important common factors. If Σ appears to deviate significantly from a diagonal matrix, then a factor model can be entertained and the factor analyst would like to estimate the factor loadings ℓ_{ij} and the specific variances ψ_i. There are two popular methods of parameter estimation, the *principal component method* and the *maximum likelihood method*.

Principal Component Method

Using spectral decomposition we can factor the covariance matrix in terms of its eigenvalues and eigenvectors,

$$\begin{aligned}\Sigma &= \lambda_1 \mathbf{e}_1 {\mathbf{e}_1}^t + \lambda_2 \mathbf{e}_2 {\mathbf{e}_2}^t + \cdots + \lambda_p \mathbf{e}_p {\mathbf{e}_p}^t \\ &= \mathbf{L}\mathbf{L}^t + \mathbf{0}, \quad \text{and} \quad L = \sqrt{\lambda_1}\mathbf{e}_1, \cdots, \sqrt{\lambda_p}\mathbf{e}_p\end{aligned}$$

where $(\lambda_i, \mathbf{e}_i)$ are eigenvalue-eigenvector pairs of Σ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$. This is the case where the factor analysis model has as many factors as it does variables $(m = p)$ and specific variances $\psi_i = 0$ for all i. Where this factor analysis representation of Σ is exact it is not useful as it does not allow for any variation of specific factors ε in $(\mathbf{x} - \mu) = \mathbf{L}\mathbf{F} + \varepsilon$. Allowing for specific factors we need to take a look at the first $m(< p)$ eigenvalues and eigenvectors then

$$\Sigma = \mathbf{L}_{(p\times m)}\mathbf{L}^t_{(m\times p)} + \Psi$$

In this case $L = \sqrt{\lambda_1}\mathbf{e}_1, \cdots, \sqrt{\lambda_m}\mathbf{e}_m$ where $\psi_i = \sigma_{ii} - \sum_{j=1}^{m} \ell_{ij}^2$, $i = 1, \cdots, p$.

To continue with the principal component method it is customary to use the centered $\mathbf{x}$, for example the ith row would resemble $\mathbf{x}_i - \bar{\mathbf{x}} = x_{i1} - \bar{x}_1, x_{i2} - \bar{x}_2, \cdots, x_{ip} - \bar{x}_p$, for $i = 1, \cdots, n$. In some cases the units of the variables may not correspond in size or degree and it would be desirable for the factor analyst to work with standardized variables in which case $\mathbf{z_i} = \frac{x_{i1}-\bar{x}_1}{\sqrt{s_{11}}}, \frac{x_{i2}-\bar{x}_2}{\sqrt{s_{22}}}, \cdots, \frac{x_{ip}-\bar{x}_p}{\sqrt{s_{pp}}}$, for $i = 1, \cdots, n$ and the sample covariance matrix simply becomes the sample correlation matrix $\mathbf{R}$ of the observations $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n$. Furthermore, standardization helps to avoid problems regarding one variable with large variance unreasonably influencing the determination of the factor loadings. The estimated loadings for a given factor does not change as m increases, in other words adding another factor only adds an addition eigenvalue-eigenvector pair into the loading matrix. Estimated proportion of total sample variance due to j^{th} factor is as follows.

$$\frac{\hat{\lambda}_j}{s_{11} + \cdots + s_{pp}} \quad \text{for } \mathbf{S}\,, \qquad \frac{\hat{\lambda}_j}{p} \quad \text{for } \mathbf{R}$$

Retain m factors that provides a satisfactory interpretation of the data and yields a satisfactory fit to $\mathbf{S}$ or $\mathbf{R}$.

Maximum Likelihood Method

Assume that $\mathbf{F}$ and ε are jointly normal then the observations from our model,

$$\mathbf{x} - \mu = \mathbf{LF} + \varepsilon$$

are then normal; therefore the likelihood function becomes

$$
\begin{aligned}
L(\mu,\Sigma) &= \prod_{i=1}^{n} \frac{1}{(2\pi)^p|\Sigma|} \exp\left(-\frac{1}{2}(\mathbf{x}_i-\mu)^t\Sigma^{-1}(\mathbf{x}_i-\mu)\right) \\
&= \frac{1}{(2\pi)^{np/2}|\Sigma|^{n/2}} \exp \quad -\frac{1}{2}\sum_{i=1}^{n}(\mathbf{x}_i-\mu)^t\Sigma^{-1}(\mathbf{x}_i-\mu) \\
&= \frac{1}{(2\pi)^{np/2}|\Sigma|^{n/2}} \exp \quad -\frac{1}{2}tr\left[\Sigma^{-1} \quad \sum_{i=1}^{n}(\mathbf{x}_i-\bar{\mathbf{x}})(\mathbf{x}_i-\bar{\mathbf{x}})^t + n(\bar{\mathbf{x}}-\mu)(\bar{\mathbf{x}}-\mu)^t \quad \right]\Bigg) \\
&= \frac{1}{(2\pi)^{(n-1)p/2}|\Sigma|^{(n-1)/2}} \exp \quad -\frac{1}{2}tr\left[\Sigma^{-1} \quad \sum_{i=1}^{n}(\mathbf{x}_i-\bar{\mathbf{x}})(\mathbf{x}_i-\bar{\mathbf{x}})^t \quad \right]\Bigg) \\
&\quad \times \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left(-\frac{n}{2}(\bar{\mathbf{x}}-\mu)^t\Sigma^{-1}(\bar{\mathbf{x}}-\mu)\right)
\end{aligned}
$$

where tr stands for the trace of a matrix.

This likelihood function depends on $\mathbf{L}$ and Ψ through $\Sigma = \mathbf{L}\mathbf{L}^t + \Psi$. Although this model is still not well defined due to the choices for $\mathbf{L}$ through orthogonal transformations, so it is desirable to set a uniqueness condition on $\mathbf{L}$ to make it well defined and this condition is $\mathbf{L}^t\Psi^{-1}\mathbf{L} = \Delta$ where Δ is a diagonal matrix. The estimated proportion of total sample variance due to j^{th} factor is as follows.

$$
\frac{\hat{\ell}_{1j}^2 + \hat{\ell}_{2j}^2 + \cdots + \hat{\ell}_{pj}^2}{s_{11}+s_{22}+\cdots+s_{pp}} \quad \text{for } \mathbf{S}\,, \qquad \frac{\hat{\ell}_{1j}^2 + \hat{\ell}_{2j}^2 + \cdots + \hat{\ell}_{pj}^2}{p} \quad \text{for } \mathbf{R}
$$

4.4 Factor Rotation

The solution from either method can be rotated, through an orthogonal transformation, in order to simplify the interpretation of factors. If $\hat{\mathbf{L}}$ is the $p \times m$ matrix of estimated factor loadings, obtained by either method in Section 4.3, then

$$\hat{\mathbf{L}}^* = \hat{\mathbf{L}}\mathbf{T}, \qquad \text{where} \quad \mathbf{T}\mathbf{T}^t = \mathbf{T}^t\mathbf{T} = \mathbf{I}$$

$\hat{\mathbf{L}}^*$ is a $p \times m$ matrix of *rotated* loadings. Furthermore the estimated covariance or correlation matrix remains unchanged using rotated loadings since

$$\hat{\mathbf{L}}\hat{\mathbf{L}}^t + \hat{\Psi} = \hat{\mathbf{L}}\mathbf{T}\mathbf{T}^t\hat{\mathbf{L}}^t + \hat{\Psi} = \hat{\mathbf{L}}^*\hat{\mathbf{L}}^{*t} + \hat{\Psi} = \Sigma$$

which provides an unchanged residual matrix as $\mathbf{S}_n - \hat{\mathbf{L}}\hat{\mathbf{L}}^t - \hat{\Psi} = \mathbf{S}_n - \hat{\mathbf{L}}^*\hat{\mathbf{L}}^{*t} - \hat{\Psi}$. Furthermore, the specific variances $\hat{\psi}_i$ remain untouched by the rotation thus a rotation is plausible.

There have been several types of rotation methods employed by factor analyst, but the *Varimax* method, developed by Henry Kaiser in 1958, is the most popular method. This is due to the fact that *Varimax* uses orthogonal rotations to search for a rotation such that the variance of the loadings is maximized to provide simple interpretation as each factor has a small number of large loadings and a large number of zero, or small, loadings. This tends to produce multiple group factors and after rotation, each original variable tends to be associated with one, or a small number, of factors.

Some of the other rotation methods include the *Quartimax method*, the *Equamax method*, and the *Promax method*. The *Quartimax* method tends to produce a general factor and additional smaller group factors, but in cases where strong group factors are present the *Quartimax* method will not make these factors dissipate. The goal of the *Quartimax* method is to minimize the number of factors needed to explain each variable.

The *Equamax* method specifies the orthogonal rotation in such a way that it maximizes a weighted sum of the *Varimax* and *Quartimax* criterion, reflecting a concern for simple structures within factors as well as within variables.

The *Promax* method tends to produce group factors that look like *Varimax* although this method uses an oblique rotation rather than an orthogonal rotation. An oblique rotation is a non rigid rotation of the coordinate axis and provides a minimum number of factors as long as the variables are grouped into nonoverlapping clusters, otherwise this type of rotation allows for correlations between variables. The *Promax* method is computationally quick and is commonly used with very large factorings.

4.5 Diagnostics and Adequacy

While factor analysis is still a relatively new statistical method, the way a factor analyst proceeds forward with their strategy could be unique from how another factor analyst would go about the data set. Typically in factor analysis the interest usually falls on the parameters in the model, but the estimated values of the common factors, called *factor scores*, are often used for diagnostics and future analysis. Factor scores are estimates, $\hat{\mathbf{f}}_j$, of values for the unobserved random factor vectors $\mathbf{F}_j$, where $j = 1, 2, \cdots, n$.

The Weighted Least Squares Method

Given that the mean vector μ, the factor loadings $\mathbf{L}$, and the specific variances ψ are known for the factor model $\mathbf{x} - \mu = \mathbf{LF} + \varepsilon$. Furthermore the specific factors $\varepsilon^t = [\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_p]$ are considered as errors. The sum of the squares of the errors, weighted by the reciprocal of their variances, is

$$\sum_{i=1}^{p} \frac{\varepsilon_i^2}{\psi_i} = \varepsilon^t \Psi^{-1} \varepsilon = (\mathbf{x} - \mu - \mathbf{Lf})^t \Psi^{-1} (\mathbf{x} - \mu - \mathbf{Lf})$$

choosing the estimates $\hat{\mathbf{f}}$ of $\mathbf{f}$ to minimize this equation is the following.

$$\hat{\mathbf{f}} = (\mathbf{L}^t \Psi^{-1} \mathbf{L})^{-1} \mathbf{L}^t \Psi^{-1} (\mathbf{x} - \mu)$$

Taking the estimates $\hat{\mathbf{L}}$,$\hat{\Psi}$, and $\hat{\mu} = \hat{\mathbf{x}}$ as the true values, the factor scores for the jth case can be obtained as

$$\hat{\mathbf{f}}_j = (\hat{\mathbf{L}}^t \hat{\Psi}^{-1} \hat{\mathbf{L}})^{-1} \hat{\mathbf{L}}^t \hat{\Psi}^{-1} (\mathbf{x}_j - \bar{\mathbf{x}})$$

When using the maximum likelihood method the estimates $\hat{\mathbf{L}}$ and $\hat{\Psi}$ must satisfy the condition, $\hat{\mathbf{L}}^t\hat{\Psi}^{-1}\hat{\mathbf{L}} = \hat{\Delta}$, a diagonal matrix. Then the following is true.

$$\begin{aligned}\hat{\mathbf{f}}_j &= (\hat{\mathbf{L}}^t\hat{\Psi}^{-1}\hat{\mathbf{L}})^{-1}\hat{\mathbf{L}}^t\hat{\Psi}^{-1}(\mathbf{x}_j - \hat{\mu}) \\ &= \hat{\Delta}^{-1}\hat{\mathbf{L}}^t\hat{\Psi}^{-1}(\mathbf{x}_j - \bar{\mathbf{x}}), \qquad j = 1, 2, \cdots, n\end{aligned}$$

The Regression Method

Starting with the original factor model $\mathbf{x} - \mu = \mathbf{LF} + \varepsilon$ treating the loadings matrix, $\mathbf{L}$, and specific variance matrix, Ψ, as known. If the common factors $\mathbf{F}$ and the specific factors ε are jointly normally distributed with means and covariances, as follows,

$$E(\mathbf{F}) = \mathbf{0}_{(m\times 1)}, \qquad Cov(\mathbf{F}) = E[\mathbf{FF}^t] = \mathbf{I}_{(m\times m)}$$

$$E(\varepsilon) = \mathbf{0}_{(p\times 1)}, \qquad Cov(\varepsilon) = E[\varepsilon\varepsilon^t] = \Psi_{(p\times p)}$$

then the linear combination $\mathbf{x} - \mu = \mathbf{LF} + \varepsilon$ has a p-variate normal distribution with mean vector $\mathbf{0}$ and variance-covariance matrix $\mathbf{LL}^t + \Psi$. In a short hand notation, $N_p(\mathbf{0}, \mathbf{LL}^t + \Psi)$, furthermore the joint distribution of $\mathbf{x} - \mu$ and $\mathbf{F}$ is $N_{m+p}(\mathbf{0}, \Sigma^*)$, where

$$\Sigma^*_{(m+p)\times(m+p)} = \begin{bmatrix} \mathbf{LL}^t + \Psi & \mathbf{L}_{(p\times m)} \\ \mathbf{L}^t_{(m\times p)} & \mathbf{I}_{(m\times m)} \end{bmatrix} = \begin{bmatrix} \Sigma_{(p\times p)} & \mathbf{L}_{(p\times m)} \\ \mathbf{L}^t_{(m\times p)} & \mathbf{I}_{(m\times m)} \end{bmatrix}$$

Furthermore, the conditional distribution of $\mathbf{F}|\mathbf{x}$ is multivariate normal with mean and covariance as

$$E(\mathbf{F}|\mathbf{x}) = \mathbf{L}^t\Sigma^{-1}(\mathbf{x} - \mu) = \mathbf{L}^t(\mathbf{LL}^t + \Psi)^{-1}(\mathbf{x} - \mu)$$

and

$$Cov(\mathbf{F}|\mathbf{x}) = \mathbf{I} - \mathbf{L}^t\Sigma^{-1}\mathbf{L} = \mathbf{I} - \mathbf{L}^t(\mathbf{LL}^t + \Psi)^{-1}\mathbf{L}$$

The quantities $\mathbf{L}^t(\mathbf{LL}^t + \Psi)^{-1}$ are the coefficients in a multivariate regression of the factors on the variables. Given any vector of observations $\mathbf{x}_j$, and using the maximum

likelihood estimates $\hat{\mathbf{L}}$ and $\hat{\Psi}$ as the true values, the jth factor score vector is given by

$$\begin{aligned}\hat{\mathbf{f}}_j &= \hat{\mathbf{L}}^t\hat{\Sigma}^{-1}(\mathbf{x}_j - \bar{\mathbf{x}}) \\ &= \hat{\mathbf{L}}^t(\hat{\mathbf{L}}\hat{\mathbf{L}}^t + \hat{\Psi})^{-1}(\mathbf{x}_j - \bar{\mathbf{x}}) \\ &= (\mathbf{I} + \hat{\mathbf{L}}^t\hat{\Psi}^{-1}\hat{\mathbf{L}})^{-1}\hat{\mathbf{L}}^t\hat{\Psi}^{-1}, \qquad j = 1, 2, \cdots, n\end{aligned}$$

The last identity $\hat{\mathbf{L}}^t(\hat{\mathbf{L}}\hat{\mathbf{L}}^t + \hat{\Psi})^{-1} = (\mathbf{I} + \hat{\mathbf{L}}^t\hat{\Psi}^{-1}\hat{\mathbf{L}})^{-1}\hat{\mathbf{L}}^t\hat{\Psi}^{-1}$ allows for the comparison between factor scores generated through the this regression method, $\hat{f}_j^R$, and those generated through the weighted least squares method, $\hat{f}_j^{LS}$. Then with this identity we obtain,

$$\hat{f}_j^{LS} = (\hat{\mathbf{L}}^t\hat{\Psi}^{-1}\hat{\mathbf{L}})^{-1}(\mathbf{I} + \hat{\mathbf{L}}^t\hat{\Psi}^{-1}\hat{\mathbf{L}})f_j^R = \ \mathbf{I} + (\hat{\mathbf{L}}^t\hat{\Psi}^{-1}\hat{\mathbf{L}})^{-1} \ \ f_j^R$$

For maximum likelihood estimates $(\hat{\mathbf{L}}^t\hat{\Psi}^{-1}\hat{\mathbf{L}})^{-1} = \hat{\Delta}^{-1}$ and if the elements are close to zero, then both the regression and generalized least squares methods will give similar factor scores.

It is common to use the original sample covariance matrix , $\mathbf{S}$, in place of $\hat{\Sigma} = \hat{\mathbf{L}}\hat{\mathbf{L}}^t + \hat{\Psi}$ to reduce effects of possible incorrect determinations of number of factors. Then the following is true.

$$\hat{\mathbf{f}}_j = \hat{\mathbf{L}}^t\mathbf{S}^{-1}(\mathbf{x}_j - \bar{\mathbf{x}}), \qquad j = 1, 2, \cdots, n$$

Chapter 5

Simulated Data Analysis

5.1 Preliminary Analysis

Now that an understanding of factor analysis has been presented, a simulated data analysis will be performed to display the objectives of this thesis. A preliminary inspection of the data will be performed followed by some methods of estimation. Both the principal component and maximum likelihood methods will be explored as well as any appropriate rotations. The outcomes of these methodologies will be compared to see if they provide similar loading groups as well as provide a form of adequacy and diagnosis.

The data set for this thesis is a contrived surveyed data set created by Dr. Karl Wuensch, a professor of Psychology at East Carolina University, and is designed for exploratory data analysis. This data set surveyed $n = 211$ consumers and their influences in buying a six-pack of beer. When the consumer was surveyed, they were asked to rate on a scale of 0 to 100 how important each of seven qualities when deciding whether or not to buy a six-pack of beer. The variables of the data set are as follows in Table 5.1.

Table 5.1: Variables of the Data Set

Variables	Description
Cost	Importance of low cost of a six-pack
Size	Importance of larger size per bottle by volume
Alcohol	Importance of high percentage of alcohol in the beer
Reputation	Importance of the reputation of the brand of beer
Color	Importance of the color of the beer
Aroma	Importance of a nice aroma of the beer
Taste	Importance of good taste to the beer

After an initial inspection of the data set it seems that a few of the consumers did not respond in whole to the survey questions, as seen in Table 5.2, and in doing so remove themselves from the data set. This removal brings the total number of consumers that

Table 5.2: Summary of the Data Set

	Cost	Size	Alcohol	Reputation	Color	Aroma	Taste
n	211	209	210	208	204	204	209
Minimum	0	0	0	0	0	0	15
Median	50	30	32.5	44	50	45	65
Mean	47.63	43.92	46.34	48.66	50.19	44.14	66.65
Maximum	100	100	100	100	100	95	100

have completed the survey down to $n = 200$. After obtaining the basic summary of all the variables of the data set, in Table 5.2, we are provided with minimum, maximum, median and mean of all the variables. It appears that every variable, with exception to Aroma and Taste, has been contributed to its full range by the consumers through this

Table 5.3: Shapiro-Wilk Tests for Normality

	Cost	Size	Alcohol	Reputation	Color	Aroma	Taste
W Statistic	0.874	0.876	0.872	0.950	0.962	0.965	0.933
p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001

survey. Although for Size, Alcohol and Reputation the medians are quite different from the means indicating skewed data. This can be confirmed through Table 5.3 as we can see that these variables do not follow from a normally distributed population. Furthermore, we are able to see that none of the variables follow from a normally distributed population and after a brief overview of the data it is clear to see that the tendencies for the value of these variables is either very high or very low.

In Table 5.4 we are provided with the correlation among these seven variables. At a quick glance we are able to see that the cost of a six-pack is highly correlated with the larger size of the beers and the high percentage alcohol content of each beer and similarly the size

Table 5.4: Correlation Among Variables

	Cost	Size	Alcohol	Reputation	Color	Aroma	Taste
Cost	1	**0.856**	**0.801**	*-0.400*	-0.052	-0.108	-0.119
Size	**0.856**	1	**0.917**	*-0.400*	0.102	0.031	-0.040
Alcohol	**0.801**	**0.917**	1	*-0.455*	0.001	-0.021	-0.069
Reputation	*-0.400*	*-0.400*	*-0.455*	1	*-0.333*	*-0.396*	*-0.384*
Color	-0.052	0.102	0.001	*-0.333*	1	**0.909**	**0.900**
Aroma	-0.108	0.031	-0.021	*-0.396*	**0.909**	1	**0.867**
Taste	-0.119	-0.040	-0.069	*-0.384*	**0.900**	**0.867**	1

of each beer is highly correlated with the alcohol content. While the color of the beer is highly correlated with the aroma of the beer as well as the taste of the beer and in similar fashion the aroma of the beer is highly correlated with the taste of the beer. The importance of the reputation of the beer has a moderately negative correlation with the rest of the variables which may indicate that a consumer interested in the reputation of a beer does not focus on any other aspect of the beer. Going into the factor analysis we should be able to pick up on the two strongly correlated groups although it is vague how the factor analysis will pair the Reputation variable. With this information a principal component factor analysis (PCFA) will be performed.

5.2 Factor Analysis

Table 5.5: Final Communality Estimates from PCFA

Cost	Size	Alcohol	Reputation	Color	Aroma	Taste	Total
0.855	0.906	0.898	0.536	0.913	0.919	0.919	5.947

Taking a look at Table 5.5 the communalities provide us with the information that we are looking at 5.947 units of common variance as we have eliminated the the specific variance portion and the proportion of common variance is $\frac{5.947}{7} = 0.850$. It is important to note the communality for the Reputation variable is considerably lower than the rest at 0.536. This variable will be something to keep an eye on for the rest of the analysis.

Table 5.6: Eigenvalues of the Correlation Matrix from PCFA

Factor	1	2	3	4	5	6	7
Eigenvalue	3.313	2.812	0.584	0.215	0.130	0.084	0.040
Cumulative	0.448	0.850	0.933	0.964	0.982	0.994	1.000

The eigenvalues of the correlation matrix found through this method are shown in Table 5.6 which helps determine the proportion of total sample variance due to the jth factor. For instance the first two factors are able to provide an 85% of cumulative proportion of total sample variance.

$$\frac{\hat{\lambda}_1 + \hat{\lambda}_2}{p} = \frac{3.313 + 2.812}{7} = 0.850$$

After these two factors the eigenvalues significantly drop and the proportion slowly increases, this gives us an idea to start with $m = 2$ factors to get an idea of how the factor loadings will turn out. Which is reinforced by the scree plot in Figure 5.1 as we can see

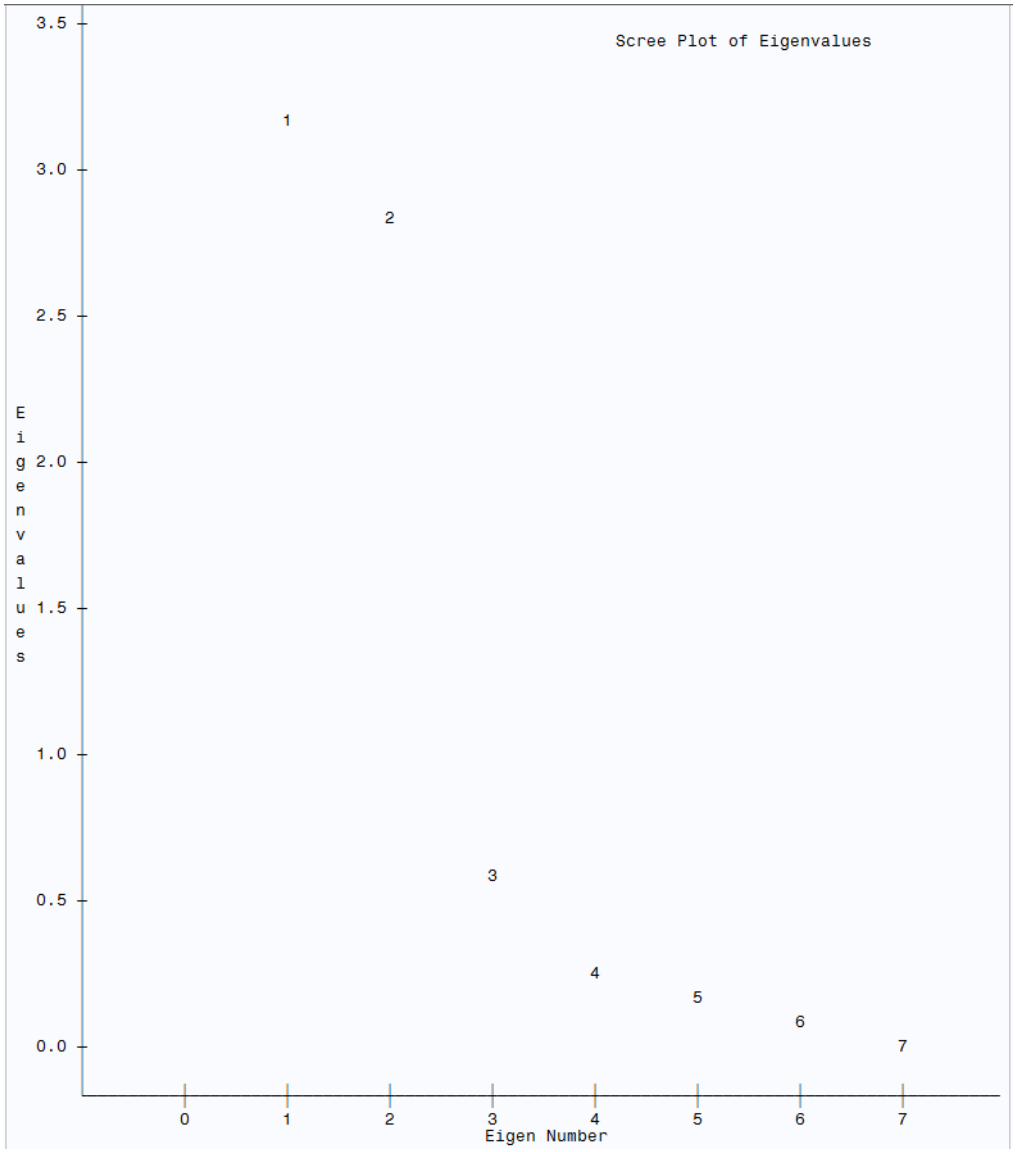

Figure 5.1: Scree Plot of Eigenvalues

a very distinct elbow in the plot occurs at the third eigenvalue. That is, the eigenvalues after $\hat{\lambda}_2$ are all relatively small and in this case it appears that two factors effectively summarize the total sample variance.

Taking a look at the factor loadings found through principal component factor analysis on Table 5.7 we are able to see how the factors have grouped the variables before any rotation has been applied. As we can see the first factor has mediocre to high loading values on all variables except for Reputation, putting the first factor to represent the idea of the reputation of the beer would be an important one, but this could be the extension of the low common variance we noticed back in Table 5.5.

Furthermore, the second factor holds especially high loading values for Cost, Size, and Alcohol, while keeping the loading values for Color, Aroma, and Taste very low. The second factor has also kept Reputation as a neutral variable with a loading value near zero. This particular grouping of variables displays a separation that indicates certain consumers to be interested solely on the ability to get drunk at low cost, which is exemplified through the cost of a six-pack, the size of each beer and alcohol content of the beer versus consumers that are interested in the quality of the beer, which is determined through the consumers interest in the color, aroma and taste of the beer. This seems to be a good separation of variables, but most of the work is being done by the second factor which is not ideal. This can be further seen when comparing the factor pattern for these two factors in Figure 5.2, the Reputation factor is on its own in quadrant three whereas the rest of the variables are in the first two quadrants. The separation displayed in this figure intuitively suggests a rotation of the factors to better fit the coordinate plane.

Table 5.7: Factor Loadings from PCFA

Rotated	Factor	Cost	Size	Alcohol	Reputation	Color	Aroma	Taste
No	1	0.567	0.677	0.640	*-0.728*	0.714	0.688	0.657
No	2	0.730	0.670	0.699	*-0.078*	-0.634	-0.667	-0.699
Yes	1	-0.096	0.026	-0.021	*-0.472*	**0.955**	**0.959**	**0.957**
Yes	2	**0.920**	**0.952**	**0.948**	*-0.560*	0.036	-0.006	-0.051

This leads us to attempt to rotate our factors in order to create a proper grouping. As seen in Figure 5.3 most of the variables have aligned themselves with an appropriate factor axis, but the Reputation variable still is completely on its own in quadrant three. This information is reinforced through Table 5.7 as we notice that the first factor has strong factor loadings with color, aroma, and taste of the beer and neutral factor loadings of the cost, the size and the alcohol content of beers, which corresponds to the consumers that are interested in the quality of the beer. The second factor has strong factor loadings with the cost of a six-pack, the size of each beer and alcohol content of the beer and neutral factor loadings with respect to color, aroma and taste of beer, which represents the consumers interested solely on their ability to get drunk at low cost.

While both of these factors have become more intuitive, there is still an issue regarding the presence of this Reputation variable that does not seem to be explained in either factor well as it is explained in equal parts due to each factor. The reputation for the beer has been a reoccurring problem variable in the analysis thus far and will continue to be watched as we go into the maximum likelihood factor analysis to see if we can verify what we have seen thus far through the principal component factor analysis method.

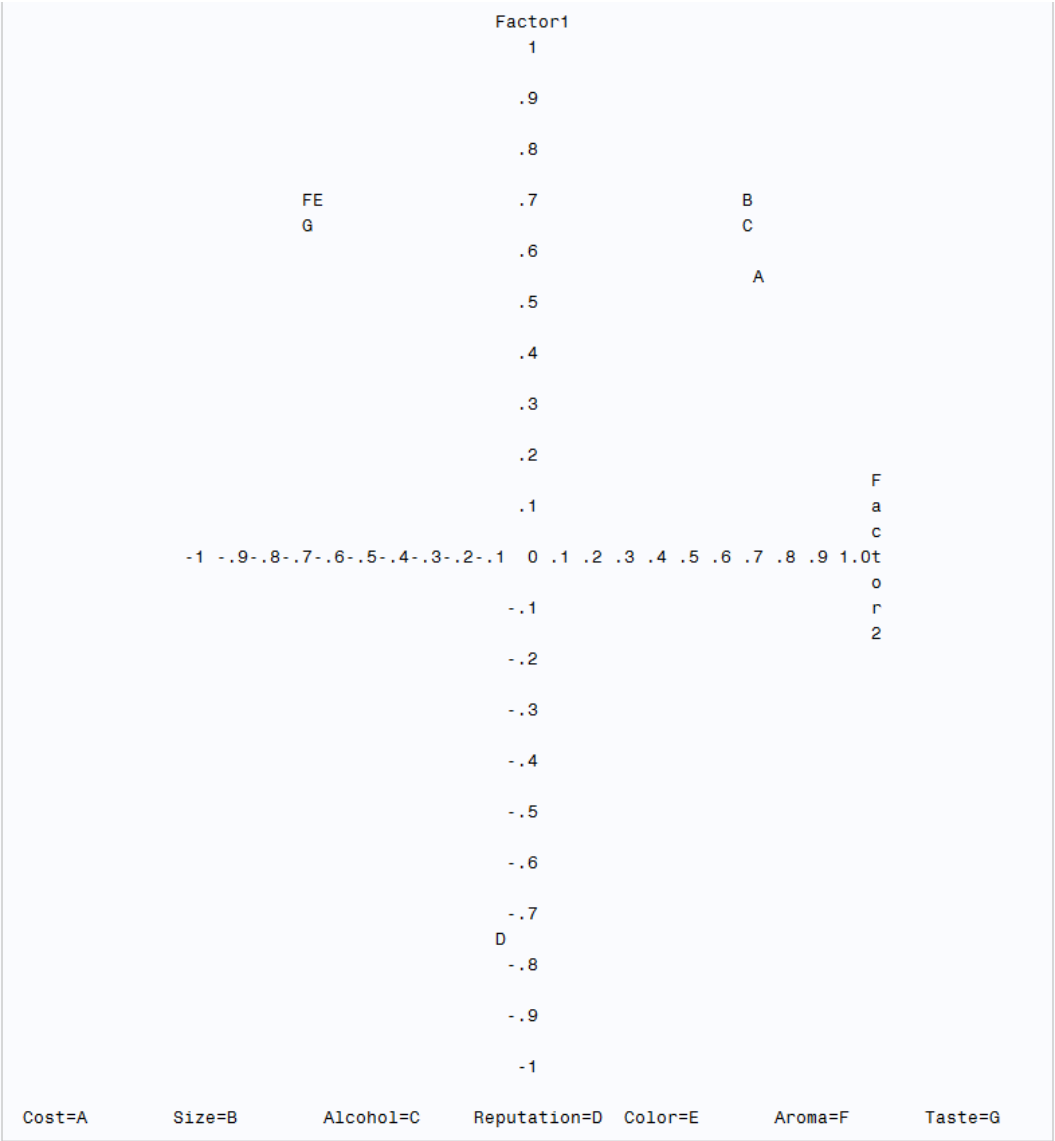

Figure 5.2: Non-rotated Factor Pattern from PCFA for Factor 1 and 2

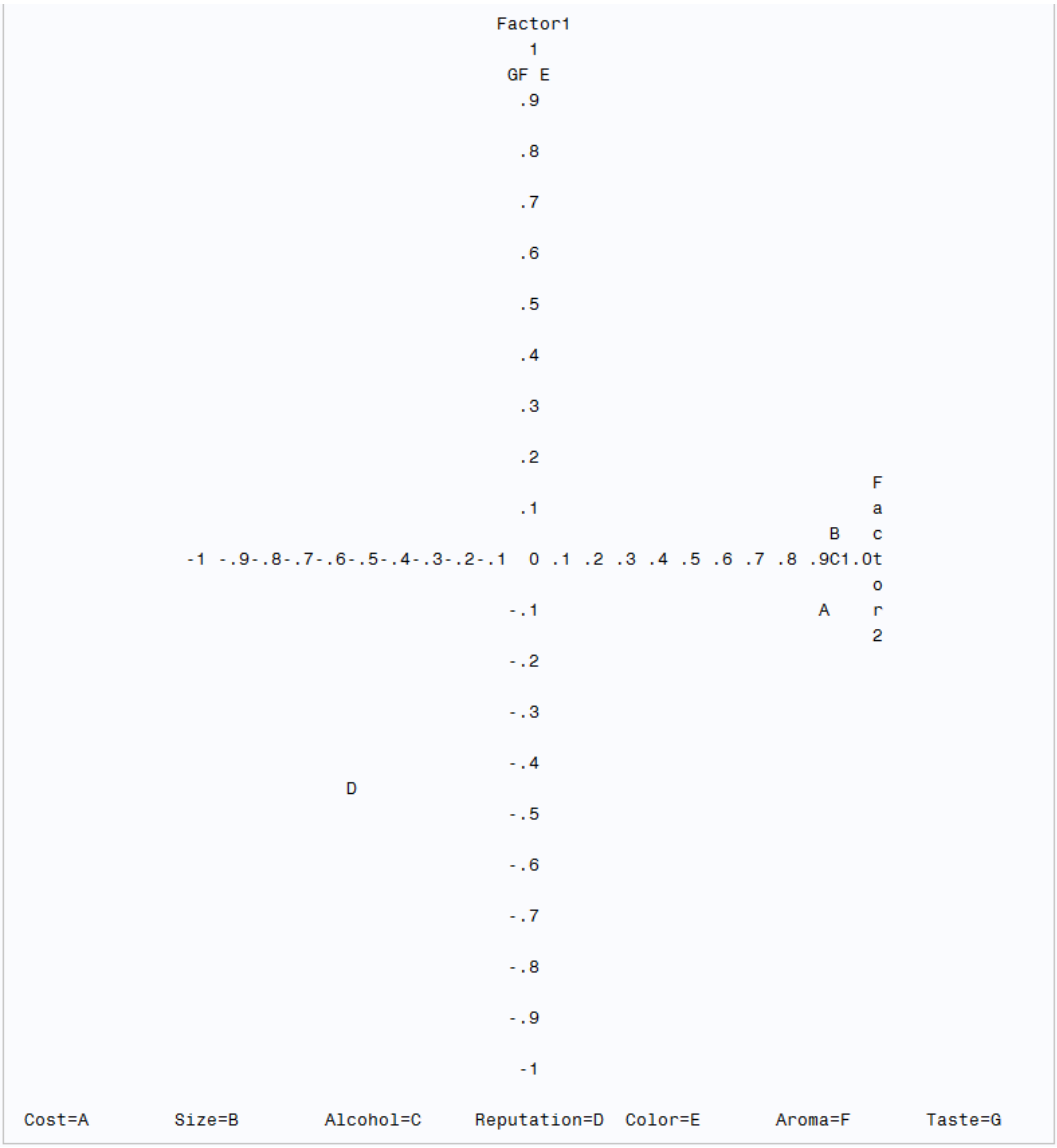

Figure 5.3: Rotated Factor Pattern from PCFA for Factor 1 and 2

Table 5.8: Final Communality Estimates from MLFA

Cost	Size	Alcohol	Reputation	Color	Aroma	Taste	Total
0.750	0.984	0.853	0.289	0.949	0.876	0.873	5.575

Table 5.9: Eigenvalues of the Correlation Matrix from MLFA

Factor	1	2	3	4	5	6	7
Eigenvalue	70.832	32.559	0.812	0.144	-0.070	-0.435	-0.451
Cumulative	0.685	1.000	1.008	1.009	1.009	1.004	1.000

As we look into the maximum likelihood factor analysis (MLFA) we take a look at Table 5.8 and the communalities provide us with a total of 5.575 units of common variance and as before these communalities are the totals without the specific variance portion. The proportion of common variance is $\frac{5.575}{7} = 0.796$; this number is lower than the number found in the principal component factor analysis (PCFA) method, but it can also be seen that the Reputation variable communality is considerably lower than its counterpart at 0.289. The Reputation variable seems to be caught up in the same issue we had in the PCFA method as it is apparent that 71.1% of its variance is unique.

The eigenvalues of the correlation matrix found through this method are shown in Table 5.9 which helps determine the proportion of total sample variance due to the jth factor. For instance the first two factors are able to provide a 100% of cumulative proportion of total sample variance. After these two factors the eigenvalues significantly drop and the proportion slowly increases, this gives us an idea to start with $m = 2$ factors as these two factors are able to provide 100% of the common variance.

Table 5.10: Factor Loadings from MLFA

Rotated	Factor	Cost	Size	Alcohol	Reputation	Color	Aroma	Taste
No	1	**0.850**	**0.991**	**0.917**	*-0.429*	0.154	0.078	0.012
No	2	-0.165	-0.041	-0.109	*-0.324*	**0.962**	**0.933**	**0.934**
Yes	1	-0.084	0.052	-0.022	*-0.363*	**0.972**	**0.936**	**0.931**
Yes	2	**0.862**	**0.991**	**0.923**	*-0.397*	0.063	-0.010	-0.076

The factor loadings found through maximum likelihood factor method on Table 5.10 provides similar information as the principal component method from Table 5.7. The factors have grouped variables in a similar fashion, although the factors have seemed to properly separate the groupings found previously, with exception to Reputation, before any rotation has been applied. This continues to push the thought that the factors may be able to assign grouped variables in the cases of a consumer interested solely on the ability to get drunk at a low cost versus a consumer that is interested in the quality of the beer that they purchase.

Applying a varimax rotation to this maximum likelihood factor method seems to streamline each factor as it sets up the first factor with very high loadings on the color, aroma and taste of the beer and neutral loadings for the cost, size and alcohol content of beer. This sets up the first factor to be associated with consumers that are interested in the quality of the beer that they purchase. On the other hand, the second factor is the opposite in the sense that there are very high loadings on the cost, size and alcohol content of the beer and neutral loadings for the color, aroma and taste of the beer. Pushing the second factor to be associated with consumers solely interested in the ability to get drunk at a low cost. But both of these factors still have a relatively moderate presence of Reputation which is not ideal in this situation.

Table 5.11: Factor Loadings with Three Factors

Rotated	Factor	Cost	Size	Alcohol	Reputation	Color	Aroma	Taste
No	1	0.567	0.677	0.640	*-0.728*	0.714	0.688	0.657
No	2	0.730	0.670	0.699	*-0.078*	-0.634	-0.667	-0.699
No	3	0.085	*0.229*	0.082	**0.678**	*0.224*	0.077	0.039
Yes	1	-0.095	0.058	-0.024	*-0.274*	**0.979**	**0.947**	**0.938**
Yes	2	**0.909**	**0.975**	**0.931**	*-0.317*	0.052	-0.031	-0.084
Yes	3	-0.166	-0.072	-0.195	**0.906**	-0.038	-0.163	-0.184

We would like to see if there is a way to neutralize the reoccurring problem of the reputation of beers not falling into a distinct factor. We have seen that two factors are able to reasonably explain the common variance and produce nice variable groupings with the exception of Reputation, so let us attempt to use three factors and witness the outcome. Taking a look at Table 5.11 at first glance the non-rotated factor loadings appear similar to the case where there were two factors in Table 5.7, but we have an additional factor that shows a large loading on the Reputation variable, 0.678. This third factor also has some small loadings associated with size and color of the beers, but still falls into the same issue as the other factor methods by containing an unclear Reputation variable among the factors.

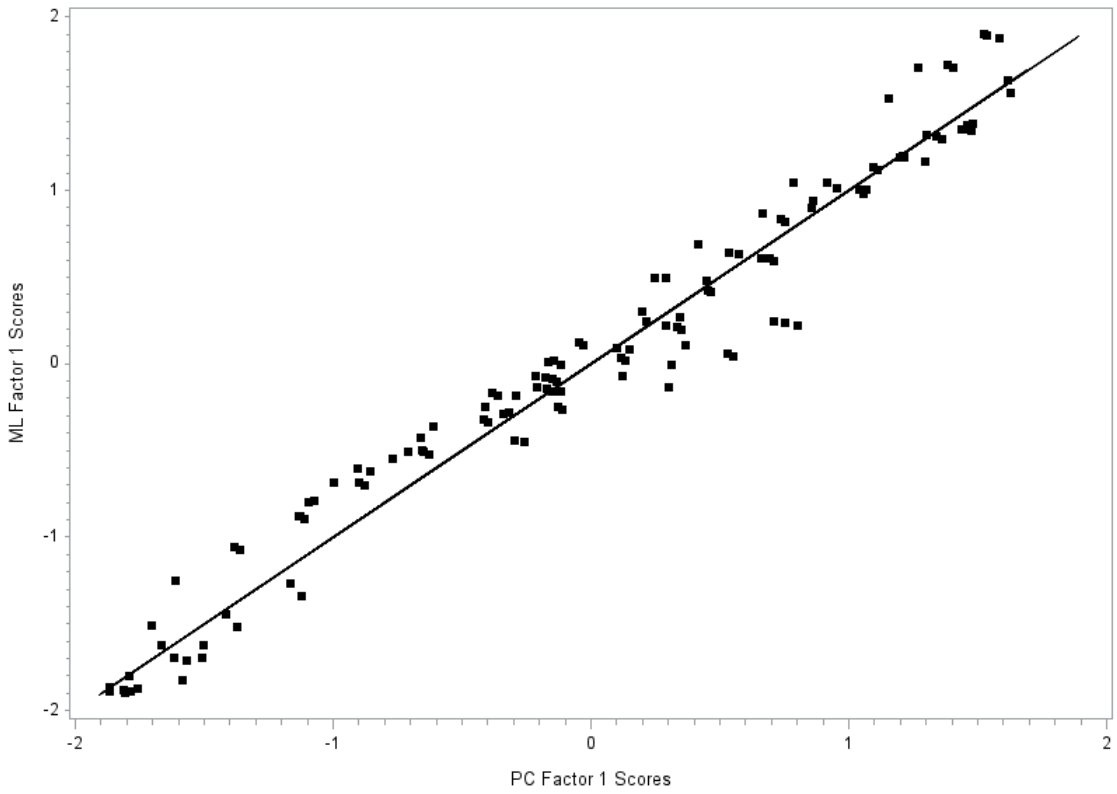

Figure 5.4: Plot of Factor Score 1 Obtained through PCFA vs MLFA

After attempting to rotate this three factor method in hopes that we are able to salvage a case where the Reputation variable would fall into a clear factor category does not quite hit the mark when we see that the rotated factors strongly agree with our two factor method in Table 5.7. However, the third factor after rotation provides a very strong loading towards Reputation, 0.906, and much lower loadings for all other variables, essentially creating a factor that is controlled by one variable, which is not ideal as some authors, Costello, recommend that a factor with fewer than three variables is generally weak and unstable. Furthermore, this can be reinforced when plotting sets of factor

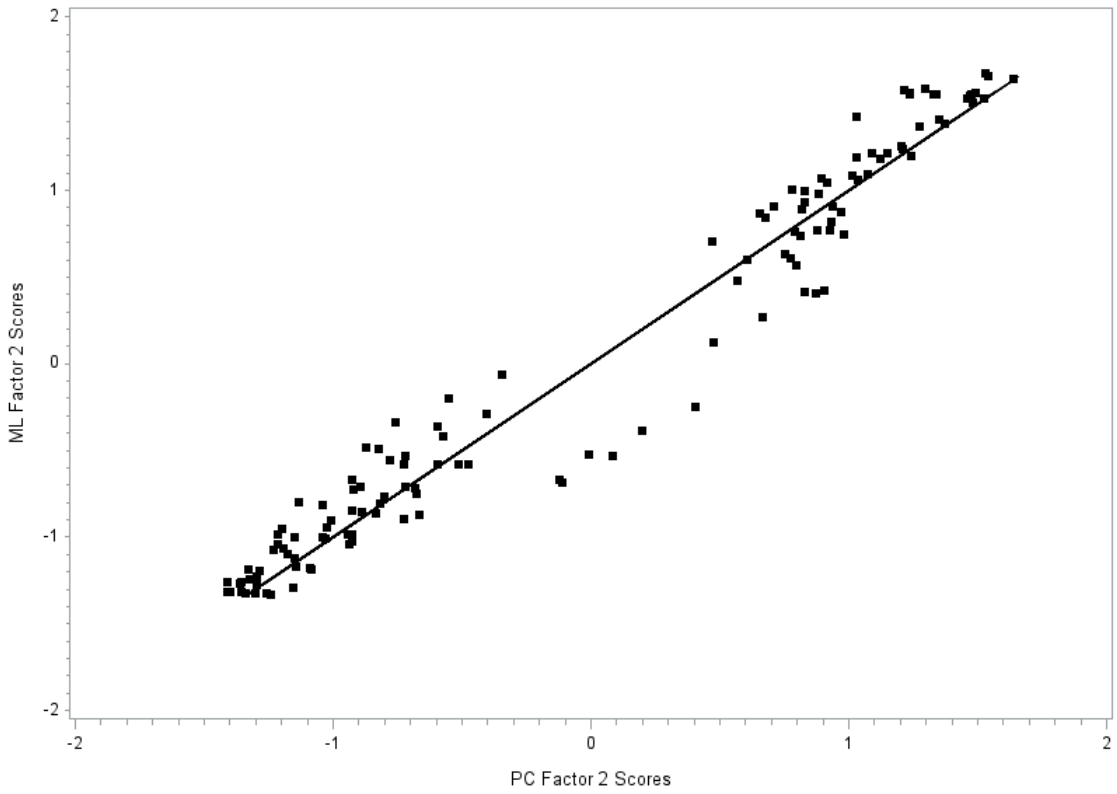

Figure 5.5: Plot of Factor Score 2 Obtained through PCFA vs MLFA

scores obtained through both principal component and maximum likelihood methods we arrive at the plots shown in Figure 5.4, Figure 5.5 and Figure 5.6. These factor scores aid as another adequacy check along the way. If the loadings of particular factor agree, the pairs of scores should cluster about the 45° line through the origin.

The sets of loadings that do not agree will produce factor scores that deviate from this pattern and is usually associated with the last factor as the amount of factors chosen is too large. In Figure 5.4 we are able to see the first factor scores and they are following a proper trend. Similarly, in Figure 5.5 we have the same general trend within reason, but in Figure 5.6 we are no longer able to hold onto the trend that follows the 45° line

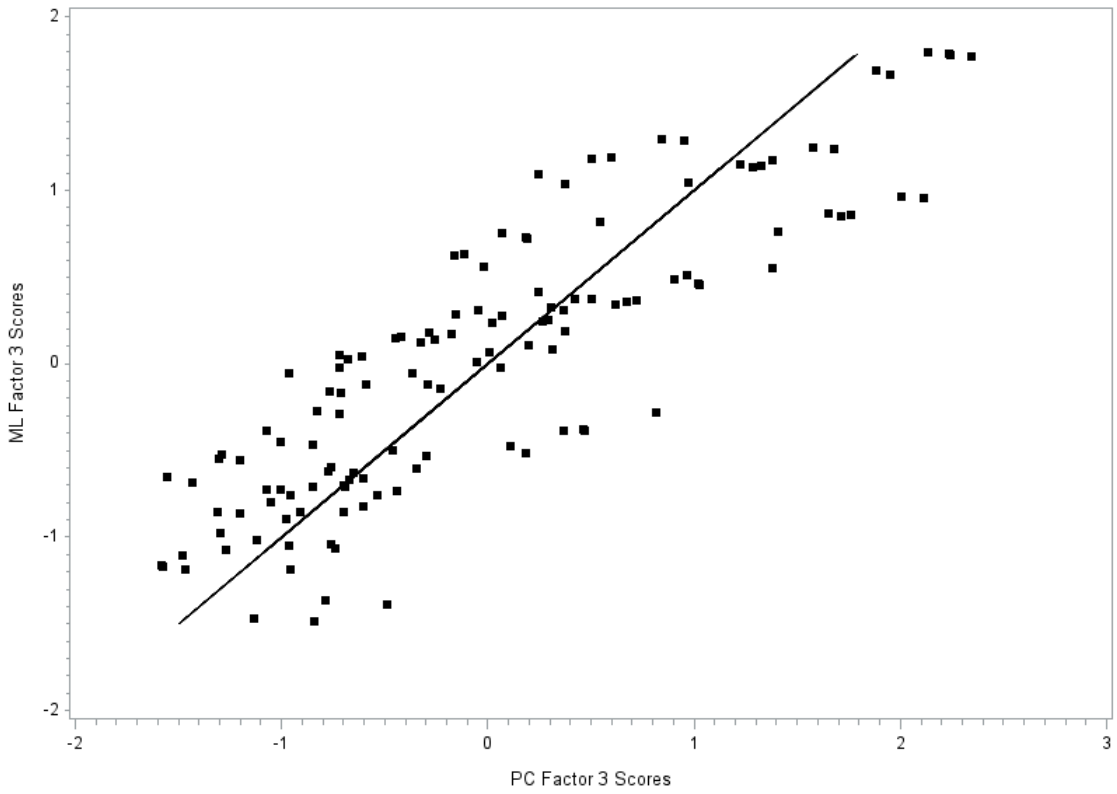

Figure 5.6: Plot of Factor Score 3 Obtained through PCFA vs MLFA

through the origin. This tells us that the last factor, the third factor, is not meaningful. This lack of meaning is reinforced by the third factor only providing a high loading for the Reputation variable. Furthermore, looking at Table 5.12 we are able to take a look at the residual correlations with uniqueness on the diagonal.

These residual correlations are reproduced through the factor solution with two factors. For an ideal solution the reproduced correlation matrix would be similar to the original correlation matrix for the data, so the elements of the residual correlation matrix are those of the difference of the original correlation matrix and the reproduced correlation matrix. These values ideally should be small, but as we can see the specific variance

Table 5.12: Residual Correlations with Uniqueness on the Diagonal

	Cost	Size	Alcohol	Reputation	Color	Aroma	Taste
Cost	0.250	0.000	-0.003	-0.094	-0.010	-0.011	0.031
Size	0.000	0.016	0.000	0.014	0.003	-0.002	-0.006
Alcohol	-0.003	0.000	0.147	-0.100	-0.021	0.015	0.032
Reputation	-0.094	0.014	-0.100	**0.711**	0.049	-0.060	-0.080
Color	-0.010	0.003	-0.021	0.049	0.051	0.000	0.001
Aroma	-0.011	-0.002	0.015	-0.060	0.000	0.124	-0.004
Taste	0.031	-0.006	0.032	-0.080	0.001	-0.004	0.127

of the Reputation variable has remained very large.

With all of the evidence gathered against the Reputation variable we will remove it from the factor analysis and keep the variable separate. We will perform another factor method without the inclusion of the Reputation variable to see if the factors behave the same way. As we can see in Table 5.13 before any rotations the factor loadings note the large trends with respect to the Cost, Size and Alcohol grouping. As well as the Color, Aroma and Taste grouping which is similar to previous methods, but this time each factor also include mediocre loading trends for the remaining variables. This is not ideal so a varimax rotation will be applied.

Table 5.13: Factor Loadings without Reputation

Rotated	Factor	Cost	Size	Alcohol	Color	Aroma	Taste
No	1	*-0.532*	*-0.438*	*-0.482*	**0.824**	**0.841**	**0.862**
No	2	**0.762**	**0.870**	**0.820**	*0.521*	*0.465*	*0.419*
Yes	1	-0.082	0.054	-0.010	**0.974**	**0.961**	**0.956**
Yes	2	**0.925**	**0.973**	**0.951**	0.042	-0.016	-0.066

Table 5.14: Final Communality Estimates without Reputation

Cost	Size	Alcohol	Color	Aroma	Taste	Total
0.863	0.949	0.904	0.950	0.923	0.919	5.508

After rotations, seen in Table 5.13, we are provided with loadings for the first factor that have very high values in color, aroma and taste of beer representing the consumers that are interested in the quality of the beer that they purchase. On the other hand, the second factor contributes very high loadings on the cost, size and alcohol content of the beer and neutral loadings for the color, aroma and taste of the beer. Pushing the second factor to be associated with consumers solely interested in the ability to get drunk at a low cost. Furthermore, in Table 5.14 the communalities provide a total of 5.508 units of common variance and the proportion of common variance is $\frac{5.508}{6} = 0.918$. Indicating that not only do these common factors provide a good representation in terms of explaining common variability, but the variables also fall into good factor groupings.

Table 5.15: Final Factor Model

		Factor 1 (Connoisseur)		Factor 2 (Cheap Drunk)
Color	=	$0.974F_1$		
Aroma	=	$0.961F_1$		
Taste	=	$0.956F_1$		
Cost	=		+	$0.925F_2$
Size	=		+	$0.973F_2$
Alcohol	=		+	$0.951F_2$

5.3 Conclusion and Future Research

Factor analysis is a multivariate statistical tool that is used in the development, refinement and evaluation of tests and measures of data that can be used in clinical and educational contexts. Factor analysis is commonly used in education, health related professions, sociology, psychology, and political science. The objective of this thesis is to provide an exploratory factor analysis procedure, and after applying several different methods of factor analysis we were able to come to the conclusion seen in Table 5.15 providing the final factor model. The first factor has very high loading values in color, aroma and taste of beer representing the consumers that are interested in the quality of the beer that they purchase representing a beer connoisseur. Furthermore, the second factor has very high loading values on the cost, size and alcohol content of the beer associating these consumers solely based on interest in the ability to get drunk at a low cost representing a cheap drunk.

After the extraction of factors has occurred the analyst has a better understanding of the material and can continue on to use the factors in other analyses, such as multiple

regression. We would replace the variables in the original data set with the appropriate factors and the corresponding factor scores would become the values for these new variables. However, it should be noted that the analyst should be careful when using factors in place of variables, while the factors help reduce the multicollinearity in an original multiple regression model the factors are a scale measure of the underlying dimensions included. These factored variables tend to create a higher R^2 value than the original variables when used in regression models as the factors were groupings of highly correlated variables. Furthermore, when using factors in regression analysis it is important to avoid using unstandardized variables and coefficients as it could lead to inadequate interpretations. Due to all the restrictions that factors will cause in a multiple regression analysis another place to turn would be Structural Equation Modeling (SEM) as it is a technique that allows for multiple dependent variables. Structural equation models consist of structural and measurement models, the measurement models are similar to factor analysis as it compares how observed values relate to unobserved values.

5.4 Data Set

"COST"	"SIZE"	"ALCOHOL"	"REPUTATION"	"COLOR"	"AROMA"	"TASTE"
90	80	70	20	50	70	60
75	95	100	50	55	40	65
10	15	20	85	40	30	50
100	70	50	30	75	60	80
20	10	25	35	30	35	45
50	100	100	30	90	75	100
5	15	15	75	20	10	25
65	30	35	80	80	60	90
95	95	100	0	80	70	95
85	80	70	40	60	50	65
0	0	20	30	80	90	100
10	25	10	100	50	40	60
80	70	50	50	40	20	50
25	35	30	40	45	30	65
5	10	15	65	50	65	85
20	5	10	40	60	50	95
70	60	70	75	10	15	25
50	15	20	50	10	5	50
75	50	95	40	0	0	40
15	10	25	30	95	80	100
80	90	100	50	20	NA	40
5	10	15	65	50	65	85

80	NA	40	80	20	20	60
0	15	0	95	55	45	65
85	75	55	45	30	10	40
15	25	20	45	50	35	70
0	0	5	60	55	70	90
10	0	0	45	65	55	100
75	65	75	70	0	5	15
55	20	25	45	0	0	55
80	80	80	50	NA	30	40
25	35	30	40	45	30	65
5	10	15	65	50	65	85
20	10	25	35	30	35	45
50	100	100	30	90	75	100
20	5	10	40	60	50	95
80	55	100	45	10	0	30
70	60	70	75	10	15	25
5	0	15	30	100	85	100
0	0	0	80	NA	50	80
95	85	75	25	40	60	50
94	82	76	24	52	76	64
80	100	100	45	45	30	55
0	5	10	85	45	35	55
100	75	55	35	65	50	70
10	0	15	40	35	40	50
55	100	100	35	80	65	90

0	5	5	70	30	20	35
10	25	10	100	50	40	60
80	70	50	50	40	20	50
65	30	35	80	80	60	90
95	95	100	0	80	70	95
10	25	10	100	50	40	60
55	20	25	80	85	65	95
100	100	100	5	70	60	85
90	85	75	45	50	40	55
0	0	10	35	85	95	100
85	80	70	40	50	40	55
0	0	10	30	80	90	100
90	80	70	20	50	70	60
75	95	100	50	55	40	65
0	15	0	100	50	40	60
77	97	100	43	55	45	67
80	70	50	50	30	10	40
15	25	20	40	45	30	65
0	0	5	65	50	65	85
55	100	100	33	97	78	100
10	0	0	40	60	50	95
70	60	70	75	0	5	15
10	15	20	85	40	30	50
65	30	35	80	80	60	90
100	70	50	30	75	60	80

50	5	10	50	0	0	40
90	80	70	20	40	60	50
75	95	100	50	45	30	55
0	5	10	85	40	30	50
100	70	50	30	65	50	70
10	0	15	35	30	35	45
70	60	70	75	10	15	25
11	14	23	85	44	35	49
50	100	100	30	80	65	90
0	5	5	75	20	10	25
55	20	25	80	80	60	90
95	95	100	0	70	60	85
80	70	50	50	40	20	50
85	80	70	40	60	50	65
25	35	30	40	45	30	65
75	60	95	40	0	0	30
100	73	54	36	76	63	87
5	0	15	30	95	80	100
80	65	75	NA	NA	NA	90
95	85	75	15	45	65	55
80	100	100	45	50	35	60
5	10	15	90	45	35	55
95	75	45	35	80	55	85
15	5	20	30	35	40	50
20	30	35	40	50	35	70

0	5	10	65	55	70	90
15	0	5	40	65	55	100
4	1	23	37	86	94	100
75	65	75	75	5	10	20
15	10	25	30	95	80	90
5	15	15	75	20	10	25
85	80	70	40	60	50	65
0	0	20	30	80	90	100
55	10	25	50	5	0	55
95	95	100	0	80	70	95
80	55	100	40	0	0	35
10	5	20	30	100	85	100
20	NA	30	50	50	NA	85
90	80	70	20	50	70	60
75	95	100	50	55	40	65
10	15	20	85	40	30	50
20	5	10	40	60	50	95
100	70	50	30	75	60	80
55	100	100	25	85	70	95
0	10	10	80	25	15	30
5	15	15	75	20	10	25
60	25	30	85	85	65	95
100	100	100	0	75	65	90
90	85	75	35	55	45	60
0	0	15	30	85	95	100

5	20	5	100	55	45	65
85	75	55	50	35	15	45
50	15	20	50	10	5	50
10	25	10	100	50	40	60
75	50	95	40	0	0	40
80	70	50	50	40	20	50
95	85	75	25	55	75	65
80	100	100	55	60	45	70
15	20	25	90	45	35	55
100	75	55	35	80	65	85
75	65	75	80	15	20	30
55	20	25	55	15	10	55
80	55	100	45	5	5	45
20	15	30	35	100	85	100
90	90	100	75	NA	NA	NA
90	85	75	45	65	55	70
5	5	25	35	85	95	100
15	30	15	100	55	45	65
85	75	55	55	45	25	55
50	15	20	50	10	5	50
25	35	30	40	45	30	65
20	10	25	35	30	35	45
50	100	100	30	90	75	100
5	15	15	75	20	10	25
5	10	15	65	50	65	85

20	5	10	40	60	50	95
70	60	70	75	10	15	25
30	40	35	45	50	35	70
10	15	20	70	55	70	90
25	10	15	45	65	55	100
25	15	30	40	35	40	50
55	100	100	35	95	80	100
10	20	20	80	25	15	30
70	35	40	85	85	65	95
15	10	25	30	95	80	100
100	100	100	5	85	75	100
85	75	65	15	45	65	55
70	90	95	45	50	35	60
5	10	15	80	35	25	45
70	45	90	35	0	0	35
10	5	20	25	90	75	95
100	100	100	0	NA	NA	100
65	30	35	80	80	60	90
95	95	100	0	80	70	95
85	80	70	40	60	50	65
0	0	20	30	80	90	100
75	65	45	45	35	15	45
90	80	70	20	50	70	60
75	95	100	50	55	40	65
10	15	20	85	40	30	50

100	70	50	30	75	60	80
20	10	25	35	30	35	45
50	100	100	30	90	75	100
5	15	15	75	20	10	25
65	30	35	80	80	60	90
95	95	100	0	80	70	95
85	80	70	40	60	50	65
0	0	20	30	80	90	100
80	90	NA	NA	NA	NA	NA
10	25	10	100	50	40	60
80	70	50	50	40	20	50
25	35	30	40	45	30	65
5	10	15	65	50	65	85
20	5	10	40	60	50	95
70	60	70	75	10	15	25
50	15	20	50	10	5	50
75	50	95	40	0	0	40
15	10	25	30	95	80	100
80	75	85	NA	20	30	40
20	30	25	35	40	25	60
0	5	10	60	45	60	80
80	75	65	35	55	45	60
14	7	21	33	34	38	47
0	0	15	25	75	85	95
5	20	5	95	45	35	55

95	65	45	25	70	55	75
15	5	20	30	25	30	40
45	95	95	25	85	70	95
15	0	5	35	55	45	90
65	55	65	70	5	10	20
45	10	15	45	5	0	45
75	50	95	40	0	0	40
15	10	25	30	95	80	100
0	10	10	70	15	5	20
60	25	30	75	75	55	85
90	90	95	0	75	65	90
20	20	15	65	NA	NA	85
50	15	20	50	10	5	50
75	50	95	40	0	0	40

5.5 SAS Code

```
*calling data set;
data beer;
    infile 'beer.txt' firstobs=2;
    input Cost Size Alcohol Reputation Color
    Aroma Taste;
run;
*displaying data set through SAS;
proc print data = beer;
run;
*calling descriptive staticstics;
proc means data = beer;
run;
*testing variables for normality;
proc univariate normal data = beer;
run;
*calling correlation between variables;
proc corr data = beer noprob;
run;
*Principal Component Factor Analysis with 2 factors
    chosen including factor pattern and scree plots;
proc factor data=beer method=prin out=beer_prin n=2
    preplot plot scree;
    var Cost Size Alcohol Reputation Color Aroma Taste;
run;
```

```
*PCFA with rotation added;
proc factor data=beer method=prin rotate=varimax
    out=beer_prin n=2 preplot plot scree;
    var Cost Size Alcohol Reputation Color Aroma Taste;
run;
*Data set with additional factor scores attached;
proc print data=beer_prin;
run;
*Plot of Factor 1 versus Factor 2;
proc gplot data=beer_prin;
    plot factor2*factor1;
run;
*Maximum Likelihood Factor Analysis with 2 factors
    chosen including residual matrix factor pattern
    and scree plots;
proc factor data=beer method=ml rotate=varimax res
    out=beer_ml n=2 preplot plot scree;
    var Cost Size Alcohol Reputation Color Aroma Taste;
run;
*Data set with additional factor scores attached;
proc print data=beer_ml;
run;
*Plot of Factor 1 versus Factor 2;
proc gplot data=beer_ml;
    plot factor2*factor1;
```

```
run;
*PCFA with 3 factors chosen including rotation and factor
    pattern and scree plots;
proc factor data=beer method=prin rotate=varimax
    out=beer_prin n=3 preplot plot scree;
    var Cost Size Alcohol Reputation Color Aroma Taste;
run;
*Refining data to only include PCFA factor scores;
data prin_beer;
set beer_prin (KEEP = factor1 factor2 factor3);
data prin_beer (RENAME=(factor1=prinf1 factor2=
    prinf2 factor3=prinf3));
set prin_beer;
*Displaying PCFA factor scores through SAS;
proc print data = prin_beer;
run;
*MLFA with 3 factors chosen including residual matrix,
    factor pattern and scree plots as well as a heywood
    restriction to restrict communalities to 1;
proc factor data=beer method=ml rotate=varimax res
    heywood out=beer_ml n=3 preplot plot scree;
    var Cost Size Alcohol Reputation Color Aroma Taste;
run;
*Refining data to only include MLFA factor scores;
data ml_beer;
```

```
set beer_ml (KEEP = factor1 factor2 factor3);
data ml_beer (RENAME=(factor1=mlf1 factor2=mlf2
    factor3=mlf3));
set ml_beer;
*Displaying MLFA factor scores through SAS;
proc print data = ml_beer;
run;
*Refining data to include both MLFA
    factor scores and PCFA factor scores;
data mlprin_beer;
merge prin_beer ml_beer;
set mlprin_beer;
*Displaying MLFA and PCFA factor scores through SAS;
proc print data = mlprin_beer;
run;
*Creating 3 plots that show factor score
    comparisons found through MLFA and PCFA;
symbol1 value='b7'x height=54pt font='Arial' c=black;
symbol2 i=join c=black;
proc gplot data=mlprin_beer;
    axis5 label=("PC Factor 1 Scores");
    axis6 label=(angle=90 "ML Factor 1 Scores");
    plot mlf1*prinf1 mlf1*mlf1/ overlay haxis=axis5
    vaxis=axis6;
run;
```

```
proc gplot data=mlprin_beer;
    axis3 label=("PC Factor 2 Scores");
    axis4 label=(angle=90 "ML Factor 2 Scores");
    plot mlf2*prinf2 mlf2*mlf2/ overlay haxis=axis3
    vaxis=axis4;
run;
proc gplot data=mlprin_beer;
    axis1 label=("PC Factor 3 Scores");
    axis2 label=(angle=90 "ML Factor 3 Scores");
    plot mlf3*prinf3 mlf3*mlf3/ overlay haxis=axis1
    vaxis=axis2;
run;
*Two factor PCFA without Reputation variable;
proc factor data=beer method=prin rotate=varimax
    res out=beer_prin n=2 preplot plot scree;
    var Cost Size Alcohol Color Aroma Taste;
run;
```

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